Accurate Approximation Formulae for Barrier Options on Stocks with Discrete Dividends

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Abstract

The stock options with discrete dividend payout can be reasonably and consistently priced when the stock price is assumed to jump down at the exdividend date and to follow lognormal price process between two exdividend dates. However, both analytical pricing formulas and efficient numerical approaches are hard to be developed under this assumption. By approximating the discrete dividend payout with stochastic continuous dividend yields, the process of stock return can be reexpressed as a combination of several Brownian motions. The approximation analytical pricing formula for the barrier option can be derived by repeatedly applying the reflection principle and Girsanov's theorem. Numerical results suggest that our formula provide accurate pricing results.
Running title: Barrier Options on Stock with Discrete Dividends

1 Introduction

Pricing options on dividend-paying stocks is a long-standing question. By assuming that the stock price follows the log-normal diffusion process, Merton (1973) derives analytical formulae for pricing vanilla options on stocks that pay non-stochastic continuous dividend yields and this approach is widely followed by Krausz (1985), Barone-Adesi and Whaley (1987), Broadie and Detemple (1996), Broadie and Detemple (1995) Shackleton and Wojakowski (2001), Chang and Shackleton (2003), and many others. However, almost all stock dividends are paid discretely rather than continuously. Pricing stock options with discrete dividend payout seems to be first investigated in Black (1975). To address the pricing

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problem, Frishling (2002) show that the stock price processes are usually modeled by three following different ways. Model 1 suggests that the stock price minus the present value of future dividends over the life of the option follows the lognormal diffusion process (see Roll (1977)). Model 2 suggests that the stock price plus the forward values of the dividends paid from today up to option maturity follows a lognormal diffusion process (see Heath and Jarrow (1988) and Musiela and Rutkowski (1997). Model 3 suggests that the stock price jumps down with the amount of dividend paid at the exdividend date, and follows lognormal price process between two exdividend dates. Frishling (2002) argues that these three models generate very different prices. He argues that only Model 3 can reflect the reality and generate consistent option prices. The other two models can produces unreasonable pricing results especially for American-style options and some exotic options. Except the aforementioned three models, Chiras and Manaster (1978) suggests that the discrete dividends can be transformed into a fixed continuous dividend yield and then apply the Merton's formula. But Dai and Lyuu (2009) suggest that the pricing results of their approach can deviate significantly from those of Model 3. However, it is difficult to develop exact analytical formulas or efficient numerical methods for pricing vanilla options with Model 3 (see Dai (2009)). Bender and Vorst (2001), Bos and Vandermark (2002), Dai (2009), and Dai and Lyuu (2009) provide approximate analytical pricing formulas or numerical methods for pricing vanilla options. But no announced papers derive analytical pricing formulae for exotic stock options with discrete dividend payout.

The barrier option is a popular exotic option whose payoff depends on whether the path of the underlying stock has reached a certain predetermined price level called barrier. The analytical pricing formula for the barrier option can be derived if the underlying stock pays no dividends or constant dividend yield (see Rubinstein and Reiner (1991)). They derive the exact pricing formula by taking advantages of the reflection principle for Brownian motion and Girsanov's theorem due to the properties that the return of lognormal stock price process is a Brownian motion with drift. However, the process of stock return is no longer a Brownian motion after paying discrete dividends under Model 3 and thus the aforementioned method can not be directly applied. To address the problem, we follow Dai and Lyuu (2009) suggestion that the probability distribution of the after-dividend-payout stock price at time τ can be approximated by the probability distribution of the stock price that pays continuous dividend yield from the last exdividend date (or option initial date) to time τ. This continuous dividend yield is derived to be a function of discrete dividends and the stock returns by Taylor expansion. The yield can be interpreted as the shift of the drift and the volatility of the stock return. Due to the Markov property of the stock price process, the process of stock return after time τ can be approximated by a Brownian motion, and the reflection principle can be used to evaluate the probability to reach the barrier after time τ. The approximation analytical formula can be derived by repeatedly applying the reflection principle to estimate the probability distribution of the stock return at maturity given the condition that the barrier is not reached. The numerical results suggest that our approximation pricing formula provides more accurate results than other existent methods.

The paper is organized as follows. The mathematical model is briefly covered in section 2. Model 4 and the corresponding pricing formulas are derived in section 3. We will first consider the single-discrete-dividend case and then extend our approach to the multiple-discrete-dividend case. Experimental results given in section 4 verify the accuracy of our pricing formulas. Section 5 concludes the paper.

2 Preliminaries

2.1 Pricing Formula for Barrier Option

Following the Black-Scholes assumption, the stock price process $S(t)$, under the risk-neutral probability $\mathbb{P}$, is given by

$$S(t) = S(0)e^{\mu t+\sigma(W(t)-W(0))}, \tag{1}$$

where $\mu \equiv r - 0.5\sigma^2$, r denotes the annual risk-free interest rate, σ denotes the volatility, and $W(t)$ denotes the standard Brownian motion under $\mathbb{P}$. Assume that a barrier stock option initiates at time 0 and matures at time T. The payoff of a up-and-out option at maturity is as follows:

$$\texttt{payoff} = \begin{cases} (\theta S(T) - \theta X)^+ & \text{if } S_{\max} < B \\ 0 & \text{if } S_{\max} \geq B \end{cases},$$

where $(A)^+$ denotes $\max(A, 0)$, $S_{\max}$ denotes the maximum stock price between time 0 to time T, B denotes the barrier, and θ equals 1 for call options and -1 for put options Our paper focuses on up-and-out call option and the extensions to other barrier options are straight forward.

In Model 3, the stock price process is assumed to jump down at an exdividend date. Specifically, if a discrete dividend D is paid at an exdividend date τ, then the stock price falls by the amount αD at time τ to avoid arbitrage. For simplicity, α is assumed to be 1 in our pricing formulas. In general, α can be less than 1 to reflect the tax benefit effect on dividend income. An $\alpha \neq 1$ poses no difficulties for modifying our pricing formulas. The underlying stock is assumed to pay n discrete dividends over the option life, where the i-th dividend c_i is paid at time t_i.

The value of a barrier option that doesn't pay any discrete dividend can be analytically solved as mentioned in Rubinstein and Reiner (1991). To derive the pricing formula, we need to know if the price process has ever hit the barrier. Since the price process has ever hit the barrier during time period $[0, T]$ if and only if the maximum price process during time period $[0, T]$ is greater than the barrier, the pricing formula of a barrier option can be derived with the following theorem:

Theorem 2.1 *Let $\tilde{W}(t) = \alpha t + W(t)$ be a Wiener process with drift term αt and $\tilde{M}(\tau) = \max_{0 \le t \le \tau} \tilde{W}(t)$ be its maximum value over a certain time period $[0, \tau]$. The joint probability distribution of $(\tilde{M}(\tau), \tilde{W}(\tau))$ is given by*

$$f_{\tilde{M}(\tau),\tilde{W}(\tau)}(m, w) = \begin{cases} \frac{2(2m-w)}{\tau\sqrt{2\pi\tau}} e^{\alpha w - \frac{1}{2}\alpha^2\tau - \frac{1}{2\tau}(2m-w)^2} & \text{if } m \ge w^+ \\ 0 & \text{otherwise.} \end{cases}$$

This theorem can be derived by applying the reflection principle and Girsanov's theorem. Rewrite $\mu t + \sigma W(t)$ in Eq.(1) as $\sigma \hat{W}(t)$ (i.e. $\hat{W}(t) \equiv \mu t/\sigma + W(t)$). Since $\hat{W}(t)$ is a Wiener process with drift term, and we define $\hat{M}(t)$ by $\hat{M}(\tau) = \max_{0 \le t \le \tau} \tilde{W}(t)$ Theorem 2.1 gives the joint density function of $(\hat{W}(t), \hat{M}(t))$. Thus, the value of an up-and-out call option can be derived as follows:

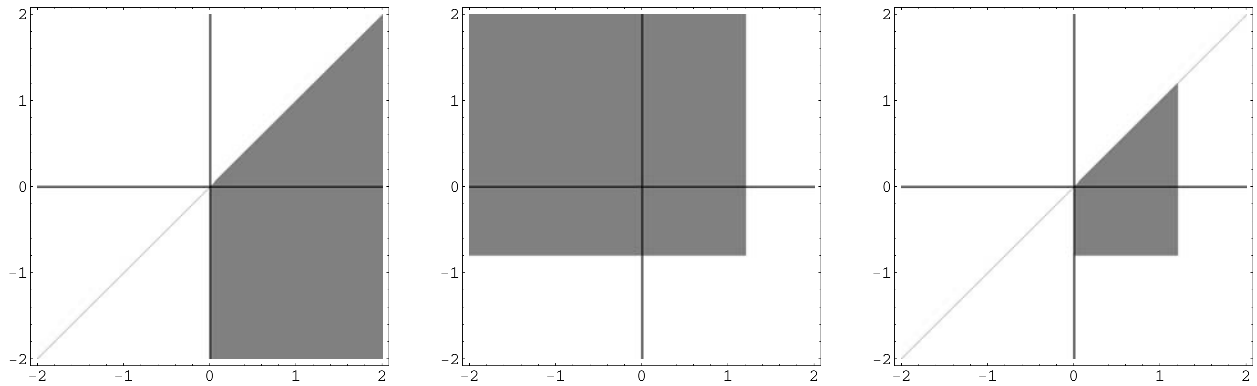
$$\begin{aligned} C &= e^{-rT} E\left\{ (S(T) - K)^+ 1_{\left\{\max\limits_{0 \le t \le T} S(t) \le B\right\}} \right\} \\ &= e^{-rT} E\left\{ (S(0)e^{\sigma\hat{W}(T)} - K) 1_{\{S(0)e^{\sigma\hat{W}(T)} \ge K, S(0)e^{\sigma\hat{M}(T)} \le B\}} \right\} \\ &= e^{-rT} E\left\{ (S(0)e^{\sigma\hat{W}(T)} - K) 1_{\{\hat{W}(T) \ge k, \hat{M}(T) \le b\}} \right\}, \end{aligned} \tag{2}$$

where k, b stand for $\frac{1}{\sigma}\log\frac{K}{S(0)}$, $\frac{1}{\sigma}\log\frac{B}{S(0)}$, respectively. Thus, applying the joint distribution given in Theorem 2.1 with $\alpha = \sigma/\mu$, we have

$$\begin{aligned} C &= e^{-rT} E\left\{ (S(0)e^{\sigma\hat{W}(T)} - K) 1_{\{\hat{W}(T) \ge k, \hat{M}(T) \le b\}} \right\} \\ &= \int_k^\infty \int_{-\infty}^b e^{-rT} \left(S(0)e^{\sigma w} - K\right) f_{\hat{M}(T),\hat{W}(T)}(m, w) dm dw && (3) \\ &= \int_k^b \int_{w^+}^b e^{-rT} \left(S(0)e^{\sigma w} - K\right) \frac{2(2m-w)}{T\sqrt{2\pi T}} e^{\alpha w - \frac{1}{2}\alpha^2 T - \frac{1}{2T}(2m-w)^2} dm dw, && (4) \end{aligned}$$

where Eq. (3) equals Eq. (4) due to the following argument: note that in the integrand $\forall m < w^+$, we have $f_{\hat{M}(T),\hat{W}(T)}(m, w) = 0$, and hence the domain of integral in Eq. (3) should be properly reformulated. $-\infty < m < b$ should be replaced by $w^+ < m < b$; on the other hand, since $w < m < b$, $k < w < \infty$ should be replaced by $k < w < b$. See Fig. 1 for the domain of integral.

To simplify Eq. (4) to a closed form pricing formula, note that only $f_{\hat{M}(T),\hat{W}(T)}(m, w)$ involve variable m, and hence $\int_{w^+}^b f_{\hat{M}(T),\hat{W}(T)}(m, w) dm$ can be evaluated first. For simplicity, we introduce the following lemma.



(a) Support (b) Integral domain (c) Taking intersection

Figure 1: Domain of double integral Eq. (4)

Panel (a) denotes the support of the density function of $f_{\hat{M}(T),\hat{W}(T)}$ (i.e. set of points (m,w) such that $f_{\hat{M}(T),\hat{W}(T)}(m,w)$ is positive), which is given by $m \geq w^+$. Panel (b) denotes the domain of integral used in Eq. (3), which is given by $(w > k) \cap (m < b)$. Panel (c) is the intersection of the the domain figured in Panel (a) and (b), which gives the domain of integral we used in Eq. (4).

Lemma 2.2

$$\int_{v^+}^{B} \frac{2(2u-v)}{\Delta\sqrt{2\pi\Delta}} e^{\alpha v - \frac{1}{2}\alpha^2\Delta - \frac{1}{2\Delta}(2u-v)^2} du = \frac{1}{\sqrt{2\pi\Delta}} e^{\alpha v - \frac{1}{2}\alpha^2\Delta - \frac{v^2}{2\Delta}} \left(1 - e^{\frac{2B(v-B)}{\Delta}}\right).$$

Proof.

$$\begin{aligned} & \int_{v^+}^{B} \frac{2(2u-v)}{\Delta\sqrt{2\pi\Delta}} e^{\alpha v - \frac{1}{2}\alpha^2\Delta - \frac{1}{2\Delta}(2u-v)^2} du = -\frac{1}{\sqrt{2\pi\Delta}} e^{\alpha v - \frac{1}{2}\alpha^2\Delta - \frac{1}{2\Delta}(2u-v)^2} \Bigg|_{u=v^+}^{u=B} \\ = & -\frac{1}{\sqrt{2\pi\Delta}} e^{\alpha v - \frac{1}{2}\alpha^2\Delta} \left(e^{-\frac{(2B-v)^2}{2\Delta}} - e^{-\frac{v^2}{2\Delta}}\right) = \frac{1}{\sqrt{2\pi\Delta}} e^{\alpha v - \frac{1}{2}\alpha^2\Delta - \frac{v^2}{2\Delta}} \left(1 - e^{\frac{2B(v-B)}{\Delta}}\right). \end{aligned}$$

□

Applying Lemma 2.2, Eq. (4) can be rewritten as

$$\begin{aligned} C &= e^{-rT} \int_k^b (S(0)e^{\sigma w} - K) \left(\int_{w^+}^{b} \frac{2(2m-w)}{T\sqrt{2\pi T}} e^{\alpha w - \frac{1}{2}\alpha^2 T - \frac{1}{2T}(2m-w)^2} dm \right) dw \\ &= e^{-rT} \int_k^b (S(0)e^{\sigma w} - K) \frac{1}{\sqrt{2\pi T}} e^{\alpha w - \frac{1}{2}\alpha^2 T - \frac{w^2}{2T}} \left(1 - e^{\frac{2b(w-b)}{T}}\right) dw \\ &= e^{-rT} \int_k^b -\frac{K}{\sqrt{2\pi T}} e^{-\frac{w^2}{2T} + \alpha w - \frac{T\alpha^2}{2}} + \frac{K}{\sqrt{2\pi T}} e^{-\frac{w^2}{2T} + \alpha w - \frac{T\alpha^2}{2} + \frac{2b(w-b)}{T}} \\ &\qquad + \frac{S'(0)}{\sqrt{2\pi T}} e^{-\frac{w^2}{2T} + \alpha w + \sigma w - \frac{T\alpha^2}{2}} - \frac{S'(0)}{\sqrt{2\pi T}} e^{-\frac{w^2}{2T} + \alpha w + \sigma w - \frac{T\alpha^2}{2} + \frac{2b(w-b)}{T}} dw, \end{aligned}$$

where the last equation is given by expanding the whole integrand immediately. In the last integral, each term of the integrand is of the form $Le^{a_2x^2+a_1x+a_0}$ (where a_0, a_1, a_2, L are constants), and hence the integral can be done applying the following identity:

$$\int_{-\infty}^{l} e^{a_2x^2+a_1x+a_0}dx = \sqrt{\frac{\pi}{-a_2}}e^{-\frac{a_1^2-4a_0a_2}{4a_2}}N\left(\frac{l-\mathbf{m}}{\mathbf{s}}\right),$$

where $\mathbf{m} = -\frac{a_1}{2a_2}$, $\mathbf{s} = \frac{1}{\sqrt{-2a_2}}$, and $N(\cdot)$ denotes the cdf of the standard normal distribution. Appendix A gives the derivation of this identity. Finally, we obtain the closed form pricing formula:

$$\begin{aligned} C = & \quad S(0)\left\{N\left(\delta_+\left(T, \frac{S(0)}{K}\right)\right) - N\left(\delta_+\left(T, \frac{S(0)}{B}\right)\right)\right\} \\ & - \; Ke^{-rT}\left\{N\left(\delta_-\left(T, \frac{S(0)}{K}\right)\right) - N\left(\delta_-\left(T, \frac{S(0)}{B}\right)\right)\right\} \\ & - \; B\left(\frac{S(0)}{B}\right)^{-\frac{2r}{\sigma^2}}\left\{N\left(\delta_+\left(T, \frac{B^2}{KS(0)}\right)\right) - N\left(\delta_+\left(T, \frac{B}{S(0)}\right)\right)\right\} \\ & + \; Ke^{-rT}\left(\frac{S(0)}{B}\right)^{-\frac{2r}{\sigma^2}+1}\left\{N\left(\delta_-\left(T, \frac{B^2}{KS(0)}\right)\right) - N\left(\delta_-\left(T, \frac{B}{S(0)}\right)\right)\right\}, \end{aligned} \tag{5}$$

where

$$\delta_\pm(\tau, s) \equiv \frac{1}{\sigma\sqrt{\tau}}\left[\log s + \left(r \pm \frac{1}{2}\sigma^2\right)\tau\right].$$

2.2 Approximate the Dividend-paying Stock Price Process

Obviously, the stock price process before the first exdividend date is as given in Eq.(1). Here we only focus on what happens after the first exdividend date.

Following the idea by Dai and Lyuu (2009), the stock price $S(t)$ between the first and second exdividend date, defined as t_1 and t_2 respectively, can be approximated by a stock price process that pays a continuous dividend yield q_1 from time 0 to time t_1 as follows:

$$\begin{aligned} S(t) &= \left[S(0)e^{\mu t_1+\sigma(W(t_1)-W(0))} - c_1\right]e^{\mu(t-t_1)+\sigma(W(t)-W(t_1))} \\ &= S(0)e^{(\mu-q_1)t_1+\sigma(W(t_1)-W(0))+\mu(t-t_1)+\sigma(W(t)-W(t_1))}, \end{aligned} \tag{6}$$

and hence

$$S(0)e^{\mu t_1+\sigma(W(t_1)-W(0))} - c_1 = S(0)e^{\mu t_1+\sigma(W(t_1)-W(0))} \cdot e^{-q_1t_1}.$$

Since q_1 is usually small, e^{-q_1t} can be approximated well by its first order Taylor expansion

$1 - q_1 t$, replacing $e^{-q_1 t_1}$ in the RHS of the above identity by $1 - q_1 t_1$ gives

$$
\begin{aligned}
S(0)e^{\mu t_1+\sigma(W(t_1)-W(0))} - c_1 &\approx S(0)e^{\mu t_1+\sigma(W(t_1)-W(0))}(1-q_1 t_1) \\
\Rightarrow -c_1 &\approx -q_1 t_1 S(0)e^{\mu t_1+\sigma(W(t_1)-W(0))} \\
\Rightarrow \quad q_1 &\approx \frac{c_1 e^{-\mu t_1}\cdot e^{-\sigma(W(t_1)-W(0))}}{t_1 S(0)} \\
\Rightarrow \quad q_1 &\approx \frac{c_1 e^{-\mu t_1}(1-\sigma(W(t_1)-W(0)))}{t_1 S(0)},
\end{aligned}
$$

where $e^x \approx 1 + x$ is used again in the last approximation. By substituting $k_1 \equiv \frac{c_1 e^{-\mu t_1}}{S(0)} + 1$, $q_1 \equiv \frac{(k_1-1)(1-\sigma(W(t_1)-W(0)))}{t_1}$ into Eq.(6), we obtain the following approximating process for Eq.(6):

$$
\begin{aligned}
S(t) &\approx S(0)e^{\mu t-(k_1-1)(1-\sigma(W(t_1)-W(0)))+\sigma(W(t_1)-W(0))+\sigma(W(t)-W(t_1))} \\
&= S(0)e^{(\mu t-k_1+1)+k_1\sigma(W(t_1)-W(0))+\sigma(W(t)-W(t_1))}
\end{aligned}
\tag{7}
$$

Note that the stock price after the second exdividend date can be recursively defined by the aforementioned method. For example, the stock price $S(t)$ between the second exdividend date t_2 and the third exdividend date t_3 can be expressed as follows:

$$
\begin{aligned}
S(t) &= \left(S(t_1)e^{\mu(t_2-t_1)+\sigma(W(t_2)-W(t_1))} - c_2\right) e^{\mu(t-t_2)+\sigma(W(t)-W(t_2))} \\
&= S(0)e^{(\mu-q_1)t_1+\sigma(W(t_1)-W(0))}e^{(\mu-q_2)(t_2-t_1)+\sigma(W(t_2)-W(t_1))}e^{\mu(t-t_2)+\sigma(W(t)-W(t_2))} \\
&\approx S(0)e^{(\mu t-k_1-k_2+2)+k_1k_2\sigma(W(t_1)-W(0))+k_2\sigma(W(t_2)-W(t_1))+\sigma(W(t)-W(t_2))},
\end{aligned}
\tag{8}
$$

where

$$
q_2 \approx \frac{(k_2-1)\left[1-k_1\sigma(W(t_1)-W(0))-\sigma(W(t_2)-W(t_1))\right]}{t_2-t_1}, \tag{9}
$$

and $k_2 \equiv \frac{c_2 e^{-\mu t_2+k_1+1}}{S(0)} - 1$.

3 Analytical Formulas

We will first derive the analytical approximating pricing formula in the single-discrete-dividend case in section 3.1. Our approach used in section 3.1 can be extended to obtain pricing formulas in general case. We will show that in section 3.2 by the same approach as in 3.1 to derive the pricing formula in the two-discrete-dividend case.

3.1 Single-discrete-dividend Case

Recall that combining Eq.(1) and (7), the stock price process can be approximated by

$$
\dot{S}(t) = \begin{cases} S(0)e^{\mu t+\sigma(W(t)-W(0))} & 0 \le t < t_1 \\ S(0)e^{(\mu t-k_1+1)+k_1\sigma(W(t_1)-W(0))+\sigma(W(t)-W(t_1))} & t_1 \le t \le T, \end{cases} \tag{10}
$$

where t_1 represents the only exdividend date. Our goal is to compute the following expectation as an approximation to call value:

$$\dot{C} \equiv e^{-rT} E\left[(\dot{S}(T) - K) 1_{\{\dot{E}_1 \cap \dot{E}_2 \cap \dot{E}_3\}}\right], \tag{11}$$

where $\dot{E}_1, \dot{E}_2$ represent the events that stock price does not hit barrier B during time period $[0, t_1)$ and $[t_1, T]$, respectively, and $\dot{E}_3$ is the event that stock price is greater than strike price at maturity date. That is, the three events $\dot{E}_1, \dot{E}_2, \dot{E}_3$ are defined by

$$\begin{aligned}
\dot{E}_1 &\equiv \left\{\dot{S}(t) < B \quad \forall \quad t \in [0, t_1)\right\}, \\
\dot{E}_2 &\equiv \left\{\dot{S}(t) < B \quad \forall \quad t \in [t_1, T]\right\}, \\
\dot{E}_3 &\equiv \left\{\dot{S}(T) > K\right\}.
\end{aligned}$$

Similarly to Eq.(3), first we rewrite the process $\dot{S}(t)$ for $t \in [t_1, T]$ as follows:

$$\begin{aligned}
\dot{S}(t) &= S(0) e^{(\mu t - k_1 + 1) + k_1 \sigma (W(t_1) - W(0)) + \sigma (W(t) - W(t_1))} \\
&= S(0) e^{(1-k_1)(1+\mu t_1) + k_1[\mu t_1 + \sigma W(t_1)] + [\mu(t - t_1) + \sigma(W(t) - W(t_1))]} \\
&= S^{'}(0) e^{k_1 \sigma \hat{W}(t_1) + \sigma \hat{W}_1(t - t_1)},
\end{aligned} \tag{12}$$

where we introduce a process $\hat{W}_1(t - t_1) \equiv \frac{\mu}{\sigma}(t - t_1) + (W(t) - W(t_1)) \ \ \forall t \in [t_1, T]$, and $S^{'}(0)$ stands for $S(0)e^{(1-k_1)(1+\mu t_1)}$ for simplicity. Putting Eq.(10) and the above identity together, with $\hat{W}(t) \equiv \frac{\mu t}{\sigma} + W(t) \ \ \forall t \in [0, t_1]$ defined as aforementioned, we obtain

$$\dot{S}(t) = \begin{cases} S(0) e^{\sigma \hat{W}(t)} & 0 \le t < t_1 \\ S^{'}(0) e^{k_1 \sigma \hat{W}(t_1) + \sigma \hat{W}_1(t - t_1)} & t_1 \le t \le T. \end{cases} \tag{13}$$

Note that the two processes $\hat{W}(t)$ for $t \in [0, t_1]$ and $\hat{W}_1(t)$ for $t \in [t_1, T]$ are independent due to the Markov property of Brownian motion. Let $\hat{M}_1(T - t_1) \equiv \max_{t_1 \le t \le T} \hat{W}_1(t - t_1)$ be the maximum value of $\hat{W}_1(t)$ over time period $[t_1, T]$. Theorem 2.1 says that the joint density functions $f_{\hat{M}(t_1), \hat{W}(t_1)}$ and $f_{\hat{M}_1(T-t_1), \hat{W}_1(T-t_1)}$ are given by

$$f_{\hat{M}(t_1), \hat{W}(t_1)}(m, w) = \begin{cases} \frac{2(2m - w)}{t_1 \sqrt{2\pi t_1}} e^{\alpha w - \frac{1}{2}\alpha^2 t_1 - \frac{1}{2t_1}(2m - w)^2} & \text{if } m \ge w^+ \\ 0 & \text{otherwise;} \end{cases} \tag{14}$$

$$f_{\hat{M}_1(T-t_1), \hat{W}_1(T-t_1)}(m_1, w_1) = \begin{cases} \frac{2(2m_1 - w_1)}{(T - t_1)\sqrt{2\pi (T - t_1)}} e^{\alpha w_1 - \frac{1}{2}\alpha^2 (T - t_1) - \frac{1}{2(T - t_1)}(2m_1 - w_1)^2} & \text{if } m_1 \ge w_1^+ \\ 0 & \text{otherwise,} \end{cases} \tag{15}$$

where $\alpha = \mu/\sigma$. For convenience, from now on we will use the symbols f_0 and f_1 to represent $f_{\hat{M}(t_1), \hat{W}(t_1)}$ and $f_{\hat{M}_1(T-t_1), \hat{W}_1(T-t_1)}$ respectively. Substituting Eq.(13) into Eq.(11), with the above joint density functions, we can compute $\dot{C}$ analytically. Note that $\dot{E}_1, \dot{E}_2, \dot{E}_3$ can be rewritten as

$$
\begin{aligned}
\dot{E}_1 &= \left\{S(0)e^{\sigma\hat{M}(t_1)} < B\right\} = \left\{\hat{M}(t_1) < b\right\}, \\
\dot{E}_2 &= \left\{S^{'}(0)e^{k_1\sigma\hat{W}(t_1)+\sigma\hat{M}_1(T-t_1)} < B\right\} = \left\{\hat{M}_1(T-t_1) < b^{'} - k_1\hat{W}(t_1)\right\}, \\
\dot{E}_3 &= \left\{S^{'}(0)e^{k_1\sigma\hat{W}(t_1)+\sigma\hat{W}_1(T-t_1)} > K\right\} = \left\{\hat{W}_1(T-t_1) > k^{'} - k_1\hat{W}(t_1)\right\},
\end{aligned}
$$

where b, $b^{'}$ and $k^{'}$ represent $\frac{1}{\sigma}\log\frac{B}{S(0)}$, $\frac{1}{\sigma}\log\frac{B}{S^{'}(0)}$ and $\frac{1}{\sigma}\log\frac{K}{S^{'}(0)}$ for simplicity. Thus, the analytical pricing formula can be derived by law of iterated expectation as follows:

$$
\begin{aligned}
\dot{C} &= e^{-rT}E\left[E\left[(\dot{S}(T)-K)1_{\{E_1\cap E_2\cap E_3\}}\middle|\hat{W}(t_1),\hat{M}(t_1)\right]\right] \\
&= e^{-rT}\int_{-\infty}^{\infty}\int_{-\infty}^{b}\int_{k^{'}-k_1w}^{\infty}\int_{-\infty}^{b^{'}-k_1w}\left(S^{'}(0)e^{k_1\sigma w+\sigma w_1}-K\right)f_1(m_1,w_1)f_0(m,w)dm_1dw_1dmdw \\
&= e^{-rT}\int_{-\infty}^{b}\int_{w^+}^{b}\int_{k^{'}-k_1w}^{b^{'}-k_1w}\int_{w_1^+}^{b^{'}-k_1w}\left(S^{'}(0)e^{k_1\sigma w+\sigma w_1}-K\right) \\
&\qquad \cdot\frac{2(2m_1-w_1)}{(T-t_1)\sqrt{2\pi(T-t_1)}}e^{\alpha w_1-\frac{1}{2}\alpha^2(T-t_1)-\frac{1}{2(T-t_1)}(2m_1-w_1)^2} \\
&\qquad \cdot\frac{2(2m-w)}{t_1\sqrt{2\pi t_1}}e^{\alpha w-\frac{1}{2}\alpha^2t_1-\frac{1}{2t_1}(2m-w)^2}dm_1dw_1dmdw. \qquad (16)
\end{aligned}
$$

Recall that in order to simplify Eq. (4), it is convenient to evaluate $\int_{w^+}^{b} f_{\hat{W}(t),\hat{M}(t)}(m,w)dm$ first. Similarly, to simplify Eq. (16) to a closed form pricing formula like Eq. (5), we start from $\int_{w^+}^{b} f_0(m,w)dm$ and $\int_{w_1^+}^{b^{'}-k_1w} f_1(m_1,w_1)$. Note that by Eq. (16), we have

$$
\begin{aligned}
\dot{C} &= e^{-rT}\int_{-\infty}^{b}\int_{k^{'}-k_1w}^{b^{'}-k_1w}\left(S^{'}(0)e^{k_1\sigma w+\sigma w_1}-K\right) \\
&\qquad \cdot\left(\int_{w_1^+}^{b^{'}-k_1w}\frac{2(2m_1-w_1)}{(T-t_1)\sqrt{2\pi(T-t_1)}}e^{\alpha w_1-\frac{1}{2}\alpha^2(T-t_1)-\frac{1}{2(T-t_1)}(2m_1-w_1)^2}dm_1\right) \\
&\qquad \cdot\left(\int_{w^+}^{b}\frac{2(2m-w)}{t_1\sqrt{2\pi t_1}}e^{\alpha w-\frac{1}{2}\alpha^2t_1-\frac{1}{2t_1}(2m-w)^2}dm\right)dw_1dw.
\end{aligned}
$$

By applying lemma 2.2 twice and the change of variable

$$
\begin{cases}
x = w_1 + k_1w \\
y = w,
\end{cases}
$$

we have [1]

$$
\begin{aligned}
\dot{C} &= e^{-rT}\int_{-\infty}^{b}\int_{k'}^{b'}\left(S'(0)e^{\sigma x}-K\right)\\
&\quad\cdot\frac{1}{\sqrt{2\pi(T-t_1)}}e^{\alpha(x-k_1y)-\frac{1}{2}\alpha^2(T-t_1)-\frac{(x-k_1y)^2}{2(T-t_1)}}\left(1-e^{\frac{2\left(b'-k_1y\right)\left(x-k_1y-\left(b'-k_1y\right)\right)}{T-t_1}}\right)\\
&\quad\cdot\frac{1}{\sqrt{2\pi t_1}}e^{\alpha y-\frac{1}{2}\alpha^2 t_1-\frac{y^2}{2t_1}}\left(1-e^{\frac{2b(y-b)}{t_1}}\right)dxdy.\\
&= e^{-rT}\int_{-\infty}^{b}\int_{k'}^{b'} I(1)+I(2)+I(3)+I(4)+I(5)+I(6)+I(7)+I(8)dxdy, \quad (17)
\end{aligned}
$$

where the integrands $I(1), I(2), \cdots, I(8)$, by immediately expanding whole the integrand, are given by

$$
\begin{aligned}
I(1) &= -\frac{K}{2\pi\sqrt{(T-t_1)\,t_1}}e^{-\frac{y^2}{2t_1}+\alpha y+\alpha(x-yk_1)-\frac{1}{2}\alpha^2(T-t_1)-\frac{\alpha^2 t_1}{2}-\frac{(x-yk_1)^2}{2(T-t_1)}},\\
I(2) &= \frac{K}{2\pi\sqrt{(T-t_1)\,t_1}}e^{-\frac{y^2}{2t_1}+\alpha y+\alpha(x-yk_1)-\frac{1}{2}\alpha^2(T-t_1)-\frac{\alpha^2 t_1}{2}-\frac{(x-yk_1)^2}{2(T-t_1)}+\frac{2(x-b')\left(b'-yk_1\right)}{T-t_1}},\\
I(3) &= \frac{K}{2\pi\sqrt{(T-t_1)\,t_1}}e^{-\frac{y^2}{2t_1}+\alpha y+\alpha(x-yk_1)-\frac{1}{2}\alpha^2(T-t_1)-\frac{\alpha^2 t_1}{2}-\frac{(x-yk_1)^2}{2(T-t_1)}+\frac{2b(y-b)}{t_1}},\\
I(4) &= -\frac{K}{2\pi\sqrt{(T-t_1)\,t_1}}e^{-\frac{y^2}{2t_1}+\alpha y+\alpha(x-yk_1)-\frac{1}{2}\alpha^2(T-t_1)-\frac{\alpha^2 t_1}{2}-\frac{(x-yk_1)^2}{2(T-t_1)}+\frac{2(x-b')\left(b'-yk_1\right)}{T-t_1}+\frac{2b(y-b)}{t_1}},\\
I(5) &= \frac{S'(0)}{2\pi\sqrt{(T-t_1)\,t_1}}e^{-\frac{y^2}{2t_1}+\alpha y+x\sigma+\alpha(x-yk_1)-\frac{1}{2}\alpha^2(T-t_1)-\frac{\alpha^2 t_1}{2}-\frac{(x-yk_1)^2}{2(T-t_1)}},\\
I(6) &= -\frac{S'(0)}{2\pi\sqrt{(T-t_1)\,t_1}}e^{-\frac{y^2}{2t_1}+\alpha y+x\sigma+\alpha(x-yk_1)-\frac{1}{2}\alpha^2(T-t_1)-\frac{\alpha^2 t_1}{2}-\frac{(x-yk_1)^2}{2(T-t_1)}+\frac{2(x-b')\left(b'-yk_1\right)}{T-t_1}},\\
I(7) &= -\frac{S'(0)}{2\pi\sqrt{(T-t_1)\,t_1}}e^{-\frac{y^2}{2t_1}+\alpha y+x\sigma+\alpha(x-yk_1)-\frac{1}{2}\alpha^2(T-t_1)-\frac{\alpha^2 t_1}{2}-\frac{(x-yk_1)^2}{2(T-t_1)}+\frac{2b(y-b)}{t_1}},\\
I(8) &= \frac{S'(0)}{2\pi\sqrt{(T-t_1)\,t_1}}e^{-\frac{y^2}{2t_1}+\alpha y+x\sigma+\alpha(x-yk_1)-\frac{1}{2}\alpha^2(T-t_1)-\frac{\alpha^2 t_1}{2}-\frac{(x-yk_1)^2}{2(T-t_1)}+\frac{2(x-b')\left(b'-yk_1\right)}{T-t_1}+\frac{2b(y-b)}{t_1}}.
\end{aligned}
$$

Since each of the integrands $I(1), I(2), \cdots, I(8)$ is a quadratic form of x, y taking exponential and multiplied by a constant, the double integral of each integrand can be expressed in terms of the cdf of a bivariate normal distribution. Like single-variate case, the cdf of a bivariate normal distribution can be approximated efficiently and accurately by a number of numerical schemes [2]. and hence all the double integrals of $I(1), I(2), \cdots, I(8)$ can be evaluated. To Simplify these double integrals, we introduce the following formula:

[1]Note that the Jacobian determinant $\frac{\partial(w_1,w)}{\partial(x,y)} = 1$.

[2]See, for example, Hull (2003)

Lemma 3.1 *Let $F_{X_1,X_2}(a,b,\Sigma)$ be the cdf of a bivariate normal distributed random vector (X_1,X_2). That is*

$$\begin{aligned} F(a,b,\Sigma) &\equiv \int_{-\infty}^{a}\int_{-\infty}^{b} f_{X_1,X_2}(x_1,x_2)dx_1dx_2 \\ &= \int_{-\infty}^{a}\int_{-\infty}^{b} \frac{1}{2\pi\sqrt{|\Sigma|}} e^{-\frac{1}{2}\mathbf{x}^T\Sigma\mathbf{x}} dx_1dx_2 \end{aligned}$$

where $\mathbf{x}$ and Σ denote $(x_1,x_2)^T$ and the covariance matrix, respectively, and both X_1 and X_2 have mean 0 and variance 1. The following double integral can be expressed in terms of F_{X_1,X_2} as:

$$\begin{aligned} & G(p,q,a_1,a_2,a_3,a_4,a_5,a_6) \\ \equiv\; & \int_{-\infty}^{q}\int_{-\infty}^{p} e^{a_1x^2+a_2xy+a_3y^2+a_4x+a_5y+a_6} dxdy \\ =\; & \frac{2\pi}{\sqrt{\Delta}} \exp\left(a_6 + \frac{a_2a_4a_5 - a_3a_4^2 - a_1a_5^2}{\Delta}\right) F_{Y_1,Y_2}\left(\frac{\sqrt{\Delta}p + \frac{2a_3a_4-a_2a_5}{\sqrt{\Delta}}}{\sqrt{-2a_3}}, \frac{\sqrt{\Delta}q + \frac{2a_1a_5-a_2a_4}{\sqrt{\Delta}}}{\sqrt{-2a_1}}, \Sigma\right), \end{aligned}$$

where $\Delta = 4a_1a_3 - a_2^2$, and Σ is given by

$$\begin{pmatrix} 1 & \frac{a_2}{2\sqrt{a_1a_3}} \\ \frac{a_2}{2\sqrt{a_1a_3}} & 1 \end{pmatrix}.$$

See Appendix A for the proof of the above lemma. With lemma 3.1, the double integrals of $I(1), I(2), \cdots, I(8)$ can be evaluate immediately. For example, the integral of $I(1)$ is given by

$$
\begin{aligned}
\int_{-\infty}^{b}\int_{k'}^{b'} I(1)dxdy &= \int_{-\infty}^{b}\int_{k'}^{b'} -\frac{K}{2\pi\sqrt{(T-t_1)\,t_1}} e^{-\frac{y^2}{2t_1}+\alpha y+\alpha(x-yk_1)-\frac{1}{2}\alpha^2(T-t_1)-\frac{\alpha^2 t_1}{2}-\frac{(x-yk_1)^2}{2(T-t_1)}} dxdy \\
= & -\frac{K}{2\pi\sqrt{(T-t_1)\,t_1}}\left(\int_{-\infty}^{b}\int_{-\infty}^{b'} e^{-\frac{1}{2(T-t_1)}x^2+\frac{k_1}{T-t_1}xy-\left(\frac{k_1^2}{2(T-t_1)}+\frac{1}{2t_1}\right)y^2+\alpha x+(\alpha-\alpha k_1)y-\frac{T\alpha^2}{2}} dxdy \right.\\
& \left. -\int_{-\infty}^{b}\int_{-\infty}^{k'} e^{-\frac{1}{2(T-t_1)}x^2+\frac{k_1}{T-t_1}xy-\left(\frac{k_1^2}{2(T-t_1)}+\frac{1}{2t_1}\right)y^2+\alpha x+(\alpha-\alpha k_1)y-\frac{T\alpha^2}{2}} dxdy\right) \\
= & -\frac{K}{2\pi\sqrt{(T-t_1)\,t_1}}\frac{2\pi}{\sqrt{\Delta(1)}}\exp\left(a_6(1)+\frac{a_2(1)a_4(1)a_5(1)-a_3(1)a_4(1)^2-a_1(1)a_5(1)^2}{\Delta(1)}\right) \\
& \cdot\left[F_{Y_1,Y_2}\left(\frac{\sqrt{\Delta(1)}b'+\frac{2a_3(1)a_4(1)-a_2(1)a_5(1)}{\sqrt{\Delta(1)}}}{\sqrt{-2a_3(1)}}, \frac{\sqrt{\Delta(1)}b+\frac{2a_1(1)a_5(1)-a_2(1)a_4(1)}{\sqrt{\Delta(1)}}}{\sqrt{-2a_1(1)}}, \Sigma(1)\right)\right. \\
& \left. -F_{Y_1,Y_2}\left(\frac{\sqrt{\Delta(1)}k'+\frac{2a_3(1)a_4(1)-a_2(1)a_5(1)}{\sqrt{\Delta(1)}}}{\sqrt{-2a_3(1)}}, \frac{\sqrt{\Delta(1)}b+\frac{2a_1(1)a_5(1)-a_2(1)a_4(1)}{\sqrt{\Delta(1)}}}{\sqrt{-2a_1(1)}}, \Sigma(1)\right)\right] \\
= & D(1)[G(b', b, a_1(1), a_2(1), a_3(1), a_4(1), a_5(1), a_6(1)) \\
& -G(k', b, a_1(1), a_2(1), a_3(1), a_4(1), a_5(1), a_6(1))],
\end{aligned}
$$

where

$$
\begin{aligned}
a_1(1) &= -\frac{1}{2\,(T-t_1)}, \\
a_2(1) &= \frac{k_1}{T-t_1}, \\
a_3(1) &= -\left(\frac{k_1^2}{2\,(T-t_1)}+\frac{1}{2t_1}\right), \\
a_4(1) &= \alpha, \\
a_5(1) &= (\alpha-\alpha k_1), \\
a_6(1) &= -\frac{T\alpha^2}{2}, \\
\Delta(1) &= 4a_1(1)a_3(1)-a_2(1)^2, \\
\Sigma(1) &= \begin{pmatrix} 1 & \frac{a_2(1)}{2\sqrt{a_1(1)a_3(1)}} \\ \frac{a_2(1)}{2\sqrt{a_1(1)a_3(1)}} & 1 \end{pmatrix}, \\
D(1) &= -\frac{K}{2\pi\sqrt{(T-t_1)\,t_1}}.
\end{aligned}
$$

Similarly, we have

$$\begin{aligned}\int_{-\infty}^{b}\int_{k'}^{b'} I(i)dxdy &= D(i)[G(b', b, a_1(i), a_2(i), a_3(i), a_4(i), a_5(i), a_6(i)) \\ &\quad -G(k', b, a_1(i), a_2(i), a_3(i), a_4(i), a_5(i), a_6(i))], \quad \forall i = 2, 3, \cdots, 8,\end{aligned}$$

where $a_1(i) = a_1(1)$, $a_3(i) = a_3(1)$, $a_2(i) = (-1)^{i+1}a_2(1)$ $\forall i = 2, 3, \cdots, 8$, and $D(i), a_4(i), a_5(i), a_6(i)$ are given by the following table:

i	$D(i)$	$a_4(i)$	$a_5(i)$	$a_6(i)$
2	$\frac{K}{2\pi\sqrt{(T-t_1)t_1}}$	$\frac{2b'}{T-t_1}+\alpha$	$\alpha+k_1\left(\frac{2b'}{T-t_1}-\alpha\right)$	$-\frac{4b'^2+T^2\alpha^2-T\alpha^2t_1}{2T-2t_1}$
3	$\frac{K}{2\pi\sqrt{(T-t_1)t_1}}$	α	$\frac{2b}{t_1}+\alpha-\alpha k_1$	$-\frac{2b^2}{t_1}-\frac{T\alpha^2}{2}$
4	$-\frac{K}{2\pi\sqrt{(T-t_1)t_1}}$	$\frac{2b'}{T-t_1}+\alpha$	$\frac{2b}{t_1}+\alpha+k_1\left(\frac{2b'}{T-t_1}-\alpha\right)$	$-\frac{2b^2}{t_1}-\frac{T\alpha^2}{2}-\frac{2b'^2}{T-t_1}$
5	$\frac{S'(0)}{2\pi\sqrt{(T-t_1)t_1}}$	$\alpha+\sigma$	$\alpha-\alpha k_1$	$-\frac{T\alpha^2}{2}$
6	$-\frac{S'(0)}{2\pi\sqrt{(T-t_1)t_1}}$	$\frac{2b'}{T-t_1}+\alpha+\sigma$	$\alpha+k_1\left(\frac{2b'}{T-t_1}-\alpha\right)$	$-\frac{4b'^2+T^2\alpha^2-T\alpha^2t_1}{2T-2t_1}$
7	$-\frac{S'(0)}{2\pi\sqrt{(T-t_1)t_1}}$	$\alpha+\sigma$	$\frac{2b}{t_1}+\alpha-\alpha k_1$	$-\frac{2b^2}{t_1}-\frac{T\alpha^2}{2}$
8	$\frac{S'(0)}{2\pi\sqrt{(T-t_1)t_1}}$	$\frac{2b'}{T-t_1}+\alpha+\sigma$	$\frac{2b}{t_1}+\alpha+k_1\left(\frac{2b'}{T-t_1}-\alpha\right)$	$-\frac{2b^2}{t_1}-\frac{T\alpha^2}{2}-\frac{2b'^2}{T-t_1}$

Thus, we obtain the pricing formula

$$\begin{aligned}\dot{C} &= e^{-rT}\int_{-\infty}^{b}\int_{k'}^{b'} I(1)+I(2)+\cdots+I(8)dxdy \\ &= e^{-rT}\sum_{i=1}^{8}\Big[D(i)[G(b', b, a_1(i), a_2(i), a_3(i), a_4(i), a_5(i), a_6(i)) \\ &\qquad -G(k', b, a_1(i), a_2(i), a_3(i), a_4(i), a_5(i), a_6(i))]\Big].\end{aligned}$$

3.2 Multi-discrete-dividend Case

The above approach can be generalized to approximate the option value in multi-discrete-dividend case. For example, the pricing formula in two-dividend case can be derived as shown below.

Following Eq.(10) and Eq.(8), in two-discrete-dividend case the stock price process can be approximated by

$$\ddot{S}(t) = \begin{cases} S(0)e^{\mu t+\sigma(W(t)-W(0))} & 0 \le t < t_1 \\ S(0)e^{(\mu t-k_1+1)+k_1\sigma(W(t_1)-W(0))+\sigma(W(t)-W(t_1))} & t_1 \le t < t_2 \\ S(0)e^{(\mu t-k_1-k_2+2)+k_1k_2\sigma(W(t_1)-W(0))+k_2\sigma(W(t_2)-W(t_1))+\sigma(W(t)-W(t_2))} & t_2 \le t \le T, \end{cases} \quad (18)$$

where t_1 and t_2 are the first and second exdividend dates. Again, we introduce the process $\hat{W}_2(t-t_2) \equiv \frac{\mu}{\sigma}(t-t_2)+(W(t)-W(t_2)) \quad \forall t \in [t_2, T]$ (and its maximum value $\hat{M}_2(T-t_2) \equiv \max_{t_2 \leq t \leq T} \hat{W}_2(t-t_2)$) and rewrite the process $\ddot{S}(t)$ for $t \in [t_2, T]$ as follows:

$$\begin{aligned}
\ddot{S}(t) &= S(0)\exp\{(\mu t - k_1 - k_2 + 2) - k_1 k_2 \mu t_1 - k_2\mu(t_2-t_1) - \mu(t-t_2) \\
&\qquad + k_1 k_2[\mu t_1 + \sigma(W(t_1) - W(0))] \\
&\qquad + k_2[\mu(t_2-t_1) + \sigma(W(t_2) - W(t_1))] \\
&\qquad + [\mu(t-t_2) + \sigma(W(t) - W(t_2))]\} \\
&= S''(0)e^{k_1 k_2 \sigma \hat{W}(t_1) + k_2\sigma\hat{W}_1(t_2-t_1) + \sigma\hat{W}_2(t-t_2)}, \qquad (19)
\end{aligned}$$

where $S''(0) \equiv S(0)e^{(\mu t - k_1 - k_2 + 2) - k_1 k_2 \mu t_1 - k_2\mu(t_2-t_1) - \mu(t-t_2)}$. Thus, together with Eq.(12), Eq.(18) can be rewritten as

$$\ddot{S}(t) = \begin{cases} S(0)e^{\sigma\hat{W}(t)} & 0 \leq t < t_1 \\ S'(0)e^{k_1\sigma\hat{W}(t_1)+\sigma\hat{W}_1(t-t_1)} & t_1 \leq t < t_2 \\ S''(0)e^{k_1k_2\sigma\hat{W}(t_1)+k_2\sigma\hat{W}_1(t_2-t_1)+\sigma\hat{W}_2(t-t_2)} & t_2 \leq t \leq T. \end{cases} \qquad (20)$$

Furthermore, applying Theorem 2.1, we have the following joint density functions:

$$f_{\hat{M}(t_1),\hat{W}(t_1)}(m,w) = \begin{cases} \frac{2(2m-w)}{t_1\sqrt{2\pi t_1}} e^{\alpha w - \frac{1}{2}\alpha^2 t_1 - \frac{1}{2t_1}(2m-w)^2} & \text{if } m \geq w^+ \\ 0 & \text{otherwise,} \end{cases} \qquad (21)$$

$$f_{\hat{M}_1(t_2-t_1),\hat{W}_1(t_2-t_1)}(m_1,w_1) = \begin{cases} \frac{2(2m_1-w_1)}{(t_2-t_1)\sqrt{2\pi(t_2-t_1)}} e^{\alpha w_1 - \frac{1}{2}\alpha^2 (t_2-t_1) - \frac{1}{2(t_2-t_1)}(2m_1-w_1)^2} & \text{if } m_1 \geq w_1^+ \\ 0 & \text{otherwise,} \end{cases} \qquad (22)$$

$$f_{\hat{M}_2(T-t_2),\hat{W}_2(T-t_2)}(m_2,w_2) = \begin{cases} \frac{2(2m_2-w_2)}{(T-t_2)\sqrt{2\pi(T-t_2)}} e^{\alpha w_2 - \frac{1}{2}\alpha^2 (T-t_2) - \frac{1}{2(T-t_2)}(2m_2-w_2)^2} & \text{if } m_2 \geq w_2^+ \\ 0 & \text{otherwise,} \end{cases} \qquad (23)$$

where $\alpha = \mu/\sigma$. For simplicity, we use the notations f_0, f_1, and f_2 to represent $f_{\hat{M}(t_1),\hat{W}(t_1)}$, $f_{\hat{M}_1(t_2-t_1),\hat{W}_1(t_2-t_1)}$, and $f_{\hat{M}_2(T-t_2),\hat{W}_2(T-t_2)}$ respectively.

Now we turn to compute the expectation

$$\ddot{C} \equiv e^{-rT}E\left[(\ddot{S}(T) - K)1_{\{\ddot{E}_1 \cap \ddot{E}_2 \cap \ddot{E}_3 \cap \ddot{E}_4\}}\right], \qquad (24)$$

where $\ddot{E}_1, \ddot{E}_2, \ddot{E}_3$ represent the events that stock price does not hit barrier B during time period $[0, t_1), [t_1, t_2)$ and $[t_2, T]$, respectively, and $\ddot{E}_4$ is the event that stock price is greater than strike price at maturity date. Similarly to single-dividend case, we rewrite $\ddot{E}_1, \ddot{E}_2, \ddot{E}_3, \ddot{E}_4$ as:

$$\begin{aligned}
\ddot{E}_1 &= \left\{S(0)e^{\sigma\hat{M}(t_1)} < B\right\} = \left\{\hat{M}(t_1) < b\right\}, \\
\ddot{E}_2 &= \left\{S^{'}(0)e^{k_1\sigma\hat{W}(t_1)+\sigma\hat{M}_1(t_2-t_1)} < B\right\} = \left\{\hat{M}_1(t_2-t_1) < b^{'} - k_1\hat{W}(t_1)\right\}, \\
\ddot{E}_3 &= \left\{S^{''}(0)e^{k_1k_2\sigma\hat{W}(t_1)+k_2\sigma\hat{W}_1(t_2-t_1)+\sigma\hat{M}_2(T-t_2)} < B\right\} \\
&= \left\{\hat{M}_2(T-t_2) < b^{''} - k_1k_2\hat{W}(t_1) - k_2\hat{W}_1(t_2-t_1)\right\}, \\
\ddot{E}_4 &= \left\{S^{''}(0)e^{k_1k_2\sigma\hat{W}(t_1)+k_2\sigma\hat{W}_1(t_2-t_1)+\sigma\hat{W}_2(T-t_2)} < K\right\} \\
&= \left\{\hat{W}_2(T-t_2) < k^{''} - k_1k_2\hat{W}(t_1) - k_2\hat{W}_1(t_2-t_1)\right\},
\end{aligned}$$

where $b, b^{'}$ are the same as aforementioned, $k^{''}$ and $b^{''}$ stand for $\frac{1}{\sigma}\log\frac{K}{S^{''}(0)}$ and $\frac{1}{\sigma}\log\frac{B}{S^{''}(0)}$, respectively. Thus, we can compute the expectation given in Eq.(24) applying the law of iterated expectation as follows:

$$\begin{aligned}
\ddot{C} &= e^{-rT}E\left[E\left[E\left[(\ddot{S}(T)-K)1_{\left\{\ddot{E}_1\cap\ddot{E}_2\cap\ddot{E}_3\cap\ddot{E}_4\right\}}\right.\right.\right. \\
&\qquad \left.\left.\left.\middle|\hat{W}(t_1),\hat{M}(t_1),\hat{W}_1(t_2-t_1),\hat{M}_1(t_2-t_1)\right]\middle|\hat{W}(t_1),\hat{M}(t_1)\right]\right] \\
&= e^{-rT}\int_{-\infty}^{\infty}\int_{-\infty}^{b}\int_{-\infty}^{\infty}\int_{-\infty}^{b^{'}-k_1w}\int_{k^{''}-k_1k_2w-k_2w_1}^{\infty}\int_{-\infty}^{b^{''}-k_1k_2w-k_2w_1} \\
&\qquad \left(S^{''}(0)e^{k_1k_2\sigma w+k_2\sigma w_1+\sigma w_2} - K\right)\cdot f_2(m_2,w_2)\cdot f_1(m_1,w_1)\cdot f_0(m,w)dm_2dw_2dm_1dw_1dmdw \\
&= e^{-rT}\int_{-\infty}^{b}\int_{w^+}^{b}\int_{-\infty}^{b^{'}-k_1w}\int_{w_1^+}^{b^{'}-k_1w}\int_{k^{''}-k_1k_2w-k_2w_1}^{b^{''}-k_1k_2w-k_2w_1}\int_{w_2^+}^{b^{''}-k_1k_2w-k_2w_1}\left(S^{''}(0)e^{k_1k_2\sigma w+k_2\sigma w_1+\sigma w_2} - K\right) \\
&\qquad \cdot\frac{2(2m_2-w_2)}{(T-t_2)\sqrt{2\pi(T-t_2)}}e^{\alpha w_2-\frac{1}{2}\alpha^2(T-t_2)-\frac{1}{2(T-t_2)}(2m_2-w_2)^2} \\
&\qquad \cdot\frac{2(2m_1-w_1)}{(t_2-t_1)\sqrt{2\pi(t_2-t_1)}}e^{\alpha w_1-\frac{1}{2}\alpha^2(t_2-t_1)-\frac{1}{2(t_2-t_1)}(2m_1-w_1)^2} \\
&\qquad \cdot\frac{2(2m-w)}{t_1\sqrt{2\pi t_1}}e^{\alpha w-\frac{1}{2}\alpha^2t_1-\frac{1}{2t_1}(2m-w)^2}dm_2dw_2dm_1dw_1dmdw.
\end{aligned}$$

To simplify this integration, first we evaluate $\int f_0(m,w)dm$, $\int f_1(m_1,w_1)dm_1$ and

$\int f_2(m_2, w_2)dm_2$ as follows:

$$
\begin{aligned}
\ddot{C} &= e^{-rT}\int_{-\infty}^{b}\int_{-\infty}^{b^{'}-k_1w}\int_{k^{''}-k_1k_2w-k_2w_1}^{b^{''}-k_1k_2w-k_2w_1}\left(S^{''}(0)e^{k_1k_2\sigma w+k_2\sigma w_1+\sigma w_2}-K\right)\\
&\quad\cdot\left(\int_{w_2^+}^{b^{''}-k_1k_2w-k_2w_1}\frac{2(2m_2-w_2)}{(T-t_2)\sqrt{2\pi(T-t_2)}}e^{\alpha w_2-\frac{1}{2}\alpha^2(T-t_2)-\frac{1}{2(T-t_2)}(2m_2-w_2)^2}dm_2\right)\\
&\quad\cdot\left(\int_{w_1^+}^{b^{'}-k_1w}\frac{2(2m_1-w_1)}{(t_2-t_1)\sqrt{2\pi(t_2-t_1)}}e^{\alpha w_1-\frac{1}{2}\alpha^2(t_2-t_1)-\frac{1}{2(t_2-t_1)}(2m_1-w_1)^2}dm_1\right)\\
&\quad\cdot\left(\int_{w^+}^{b}\frac{2(2m-w)}{t_1\sqrt{2\pi t_1}}e^{\alpha w-\frac{1}{2}\alpha^2t_1-\frac{1}{2t_1}(2m-w)^2}dm\right)dw_2dw_1dw.
\end{aligned}
$$

By applying lemma 2.2 three times and the change of variable

$$
\begin{cases}
x = w_2 + k_2w_1 + k_1k_2w\\
y = w_1 + k_1w\\
z = w
\end{cases},
$$

we have [3]

$$
\begin{aligned}
\ddot{C} &= e^{-rT}\int_{-\infty}^{b}\int_{-\infty}^{b^{'}}\int_{k^{''}}^{b^{''}}\left(S^{''}(0)e^{\sigma x}-K\right) &(25)\\
&\quad\cdot\frac{1}{\sqrt{2\pi(T-t_2)}}e^{\alpha(x-k_2y)-\frac{1}{2}\alpha^2(T-t_2)-\frac{(x-k_2y)^2}{2(T-t_2)}}\left(1-e^{\frac{2\left(b^{''}-k_2y\right)\left(x-k_2y-\left(b^{''}-k_2y\right)\right)}{(T-t_2)}}\right) &(26)\\
&\quad\cdot\frac{1}{\sqrt{2\pi(t_2-t_1)}}e^{\alpha(y-k_1z)-\frac{1}{2}\alpha^2(t_2-t_1)-\frac{(y-k_1z)^2}{2(t_2-t_1)}}\left(1-e^{\frac{2\left(b^{'}-k_1z\right)\left(y-k_1z-\left(b^{'}-k_1z\right)\right)}{(t_2-t_1)}}\right) &(27)\\
&\quad\cdot\frac{1}{\sqrt{2\pi t_1}}e^{\alpha z-\frac{1}{2}\alpha^2t_1-\frac{z^2}{2t_1}}\left(1-e^{\frac{2b(z-b)}{t_1}}\right)dxdydz = e^{-rT}\int_{-\infty}^{b}\int_{-\infty}^{b^{'}}\int_{k^{''}}^{b^{''}}\sum_{i=1}^{16}J(i)dxdydz, &(28)
\end{aligned}
$$

where, similar to $I(1), I(2), \cdots, I(8)$, we obtain $J(1), J(2), \cdots, J(16)$ by expanding the integrand, where $J(1), J(2), \cdots, J(16)$ is given in the next page. Again, all the integrands $J(1), J(2), \cdots, J(16)$ is a quadratic form taking exponential. To evaluate the triple integrals of them, we reformulate $J(1), J(2), \cdots, J(16)$ in terms of the cdf of the multivariate normal distribution by the lemma shown below. Appendix A gives the proof of this lemma.

[3]Note that the Jacobian determinant $\frac{\partial(w_2,w_1,w)}{\partial(x,y,z)}$ is still 1.

$$
\begin{aligned}
J(1) &= -\frac{K}{\sqrt{8\pi^3(T-t_2)(t_2-t_1)t_1}}e^{-\frac{z^2}{2t_1}+\alpha z+\alpha(y-zk_1)+\alpha(x-yk_2)-\frac{\alpha^2 t_1}{2}-\frac{1}{2}\alpha^2(T-t_2)-\frac{1}{2}\alpha^2(t_2-t_1)-\frac{(x-yk_2)^2}{2(T-t_2)}-\frac{(y-zk_1)^2}{2(t_2-t_1)}}, \\
J(2) &= \frac{K}{\sqrt{8\pi^3(T-t_2)(t_2-t_1)t_1}}e^{-\frac{z^2}{2t_1}+\alpha z+\alpha(y-zk_1)+\alpha(x-yk_2)-\frac{\alpha^2 t_1}{2}-\frac{1}{2}\alpha^2(T-t_2)-\frac{1}{2}\alpha^2(t_2-t_1)+\frac{2b(z-b)}{t_1}-\frac{(x-yk_2)^2}{2(T-t_2)}-\frac{(y-zk_1)^2}{2(t_2-t_1)}}, \\
J(3) &= \frac{K}{\sqrt{8\pi^3(T-t_2)(t_2-t_1)t_1}}e^{-\frac{z^2}{2t_1}+\alpha z+\alpha(y-zk_1)+\alpha(x-yk_2)-\frac{\alpha^2 t_1}{2}-\frac{1}{2}\alpha^2(T-t_2)-\frac{1}{2}\alpha^2(t_2-t_1)-\frac{(x-yk_2)^2}{2(T-t_2)}+\frac{2(x-b^{''})\left(b^{''}-yk_2\right)}{T-t_2}-\frac{(y-zk_1)^2}{2(t_2-t_1)}}, \\
J(4) &= -\frac{K}{\sqrt{8\pi^3(T-t_2)(t_2-t_1)t_1}}e^{-\frac{z^2}{2t_1}+\alpha z+\alpha(y-zk_1)+\alpha(x-yk_2)-\frac{\alpha^2 t_1}{2}-\frac{1}{2}\alpha^2(T-t_2)-\frac{1}{2}\alpha^2(t_2-t_1)+\frac{2b(z-b)}{t_1}-\frac{(x-yk_2)^2}{2(T-t_2)}+\frac{2(x-b^{''})\left(b^{''}-yk_2\right)}{T-t_2}-\frac{(y-zk_1)^2}{2(t_2-t_1)}}, \\
J(5) &= \frac{K}{\sqrt{8\pi^3(T-t_2)(t_2-t_1)t_1}}e^{-\frac{z^2}{2t_1}+\alpha z+\alpha(y-zk_1)+\alpha(x-yk_2)-\frac{\alpha^2 t_1}{2}-\frac{1}{2}\alpha^2(T-t_2)-\frac{1}{2}\alpha^2(t_2-t_1)-\frac{(x-yk_2)^2}{2(T-t_2)}-\frac{(y-zk_1)^2}{2(t_2-t_1)}+\frac{2(y-b^{'})\left(b^{'}-zk_1\right)}{t_2-t_1}}, \\
J(6) &= -\frac{K}{\sqrt{8\pi^3(T-t_2)(t_2-t_1)t_1}}e^{-\frac{z^2}{2t_1}+\alpha z+\alpha(y-zk_1)+\alpha(x-yk_2)-\frac{\alpha^2 t_1}{2}-\frac{1}{2}\alpha^2(T-t_2)-\frac{1}{2}\alpha^2(t_2-t_1)+\frac{2b(z-b)}{t_1}-\frac{(x-yk_2)^2}{2(T-t_2)}-\frac{(y-zk_1)^2}{2(t_2-t_1)}+\frac{2(y-b^{'})\left(b^{'}-zk_1\right)}{t_2-t_1}}, \\
J(7) &= -\frac{K}{\sqrt{8\pi^3(T-t_2)(t_2-t_1)t_1}}e^{-\frac{z^2}{2t_1}+\alpha z+\alpha(y-zk_1)+\alpha(x-yk_2)-\frac{\alpha^2 t_1}{2}-\frac{1}{2}\alpha^2(T-t_2)-\frac{1}{2}\alpha^2(t_2-t_1)-\frac{(x-yk_2)^2}{2(T-t_2)}+\frac{2(x-b^{''})\left(b^{''}-yk_2\right)}{T-t_2}-\frac{(y-zk_1)^2}{2(t_2-t_1)}+\frac{2(y-b^{'})\left(b^{'}-zk_1\right)}{t_2-t_1}}, \\
J(8) &= \frac{K}{\sqrt{8\pi^3(T-t_2)(t_2-t_1)t_1}}e^{-\frac{z^2}{2t_1}+\alpha z+\alpha(y-zk_1)+\alpha(x-yk_2)-\frac{\alpha^2 t_1}{2}-\frac{1}{2}\alpha^2(T-t_2)-\frac{1}{2}\alpha^2(t_2-t_1)+\frac{2b(z-b)}{t_1}-\frac{(x-yk_2)^2}{2(T-t_2)}+\frac{2(x-b^{''})\left(b^{''}-yk_2\right)}{T-t_2}-\frac{(y-zk_1)^2}{2(t_2-t_1)}+\frac{2(y-b^{'})\left(b^{'}-zk_1\right)}{t_2-t_1}}, \\
J(9) &= \frac{S^{''}(0)}{\sqrt{8\pi^3(T-t_2)(t_2-t_1)t_1}}e^{-\frac{z^2}{2t_1}+\alpha z+x\sigma+\alpha(y-zk_1)+\alpha(x-yk_2)-\frac{\alpha^2 t_1}{2}-\frac{1}{2}\alpha^2(T-t_2)-\frac{1}{2}\alpha^2(t_2-t_1)-\frac{(x-yk_2)^2}{2(T-t_2)}-\frac{(y-zk_1)^2}{2(t_2-t_1)}}, \\
J(10) &= -\frac{S^{''}(0)}{\sqrt{8\pi^3(T-t_2)(t_2-t_1)t_1}}e^{-\frac{z^2}{2t_1}+\alpha z+x\sigma+\alpha(y-zk_1)+\alpha(x-yk_2)-\frac{\alpha^2 t_1}{2}-\frac{1}{2}\alpha^2(T-t_2)-\frac{1}{2}\alpha^2(t_2-t_1)+\frac{2b(z-b)}{t_1}-\frac{(x-yk_2)^2}{2(T-t_2)}-\frac{(y-zk_1)^2}{2(t_2-t_1)}}, \\
J(11) &= -\frac{S^{''}(0)}{\sqrt{8\pi^3(T-t_2)(t_2-t_1)t_1}}e^{-\frac{z^2}{2t_1}+\alpha z+x\sigma+\alpha(y-zk_1)+\alpha(x-yk_2)-\frac{\alpha^2 t_1}{2}-\frac{1}{2}\alpha^2(T-t_2)-\frac{1}{2}\alpha^2(t_2-t_1)-\frac{(x-yk_2)^2}{2(T-t_2)}+\frac{2(x-b^{''})\left(b^{''}-yk_2\right)}{T-t_2}-\frac{(y-zk_1)^2}{2(t_2-t_1)}}, \\
J(12) &= \frac{S^{''}(0)}{\sqrt{8\pi^3(T-t_2)(t_2-t_1)t_1}}e^{-\frac{z^2}{2t_1}+\alpha z+x\sigma+\alpha(y-zk_1)+\alpha(x-yk_2)-\frac{\alpha^2 t_1}{2}-\frac{1}{2}\alpha^2(T-t_2)-\frac{1}{2}\alpha^2(t_2-t_1)+\frac{2b(z-b)}{t_1}-\frac{(x-yk_2)^2}{2(T-t_2)}+\frac{2(x-b^{''})\left(b^{''}-yk_2\right)}{T-t_2}-\frac{(y-zk_1)^2}{2(t_2-t_1)}}, \\
J(13) &= -\frac{S^{''}(0)}{\sqrt{8\pi^3(T-t_2)(t_2-t_1)t_1}}e^{-\frac{z^2}{2t_1}+\alpha z+x\sigma+\alpha(y-zk_1)+\alpha(x-yk_2)-\frac{\alpha^2 t_1}{2}-\frac{1}{2}\alpha^2(T-t_2)-\frac{1}{2}\alpha^2(t_2-t_1)-\frac{(x-yk_2)^2}{2(T-t_2)}-\frac{(y-zk_1)^2}{2(t_2-t_1)}+\frac{2(y-b^{'})\left(b^{'}-zk_1\right)}{t_2-t_1}}, \\
J(14) &= \frac{S^{''}(0)}{\sqrt{8\pi^3(T-t_2)(t_2-t_1)t_1}}e^{-\frac{z^2}{2t_1}+\alpha z+x\sigma+\alpha(y-zk_1)+\alpha(x-yk_2)-\frac{\alpha^2 t_1}{2}-\frac{1}{2}\alpha^2(T-t_2)-\frac{1}{2}\alpha^2(t_2-t_1)+\frac{2b(z-b)}{t_1}-\frac{(x-yk_2)^2}{2(T-t_2)}-\frac{(y-zk_1)^2}{2(t_2-t_1)}+\frac{2(y-b^{'})\left(b^{'}-zk_1\right)}{t_2-t_1}}, \\
J(15) &= \frac{S^{''}(0)}{\sqrt{8\pi^3(T-t_2)(t_2-t_1)t_1}}e^{-\frac{z^2}{2t_1}+\alpha z+x\sigma+\alpha(y-zk_1)+\alpha(x-yk_2)-\frac{\alpha^2 t_1}{2}-\frac{1}{2}\alpha^2(T-t_2)-\frac{1}{2}\alpha^2(t_2-t_1)-\frac{(x-yk_2)^2}{2(T-t_2)}+\frac{2(x-b^{''})\left(b^{''}-yk_2\right)}{T-t_2}-\frac{(y-zk_1)^2}{2(t_2-t_1)}+\frac{2(y-b^{'})\left(b^{'}-zk_1\right)}{t_2-t_1}}, \\
J(16) &= -\frac{S^{''}(0)}{\sqrt{8\pi^3(T-t_2)(t_2-t_1)t_1}}e^{-\frac{z^2}{2t_1}+\alpha z+x\sigma+\alpha(y-zk_1)+\alpha(x-yk_2)-\frac{\alpha^2 t_1}{2}-\frac{1}{2}\alpha^2(T-t_2)-\frac{1}{2}\alpha^2(t_2-t_1)+\frac{2b(z-b)}{t_1}-\frac{(x-yk_2)^2}{2(T-t_2)}+\frac{2(x-b^{''})\left(b^{''}-yk_2\right)}{T-t_2}-\frac{(y-zk_1)^2}{2(t_2-t_1)}+\frac{2(y-b^{'})\left(b^{'}-zk_1\right)}{t_2-t_1}}.
\end{aligned}
$$

Lemma 3.2 *Let $F_{X_1,X_2,X_3}(a,b,c,\Sigma)$ be the cdf of a multivariate normal distributed random vector (X_1,X_2,X_3). That is*

$$\begin{aligned} F(a,b,c,\Sigma) &\equiv \int_{-\infty}^{a}\int_{-\infty}^{b}\int_{-\infty}^{c} f_{X_1,X_2,X_3}(x_1,x_2,x_3)dx_1dx_2dx_3 \\ &= \int_{-\infty}^{a}\int_{-\infty}^{b}\int_{-\infty}^{c} \frac{1}{\sqrt{8\pi^3|\Sigma|}}e^{-\frac{1}{2}\mathbf{x}^T\Sigma\mathbf{x}}dx_1dx_2dx_3 \end{aligned}$$

where $\mathbf{x}$ and Σ denote $(x_1,x_2,x_3)^T$ and the covariance matrix, respectively, and each of X_1, X_2 and X_3 has mean 0 and variance 1. The following triple integral can be expressed in terms of F_{X_1,X_2,X_3} as:

$$\begin{aligned} H(p,q,r,A,B,C) &\equiv \int_{-\infty}^{r}\int_{-\infty}^{q}\int_{-\infty}^{p} e^{a_1x^2+a_2y^2+a_3z^2+a_4xy+a_5yz+a_6xz+a_7x+a_8y+a_9z+a_{10}}dxdydz \\ &= e^{C'}\sqrt{\frac{\pi^3}{|-A|}}F_{Y_1,Y_2,Y_3}\left(\frac{p_1-\mathbf{m}_1}{\mathbf{S}_{1,1}},\frac{p_2-\mathbf{m}_2}{\mathbf{S}_{2,2}},\frac{p_3-\mathbf{m}_3}{\mathbf{S}_{3,3}},\Sigma\right), \end{aligned}$$

where

$$A=\begin{pmatrix} a_1 & \frac{a_4}{2} & \frac{a_6}{2} \\ \frac{a_4}{2} & a_2 & \frac{a_5}{2} \\ \frac{a_6}{2} & \frac{a_5}{2} & a_3 \end{pmatrix}, B=\begin{pmatrix} a_7 \\ a_8 \\ a_9 \end{pmatrix}, C=a_{10}, \mathbf{S}=\begin{pmatrix} \mathbf{S}_{1,1} & 0 & 0 \\ 0 & \mathbf{S}_{2,2} & 0 \\ 0 & 0 & \mathbf{S}_{3,3} \end{pmatrix},$$

$$\begin{aligned} \mathbf{S}_{j,j} &= \sqrt{((-2A)^{-1})_{j,j}} \quad \forall j=1,2,3, \\ \mathbf{m} &= -\frac{1}{2}A^{-1}B, \\ C' &= C-\frac{1}{4}B^TA^{-1}B \\ \Sigma &= (-2\mathbf{S}A\mathbf{S})^{-1}. \end{aligned}$$

Let $A(i), B(i), C(i)$ be

$$A(i)=\begin{pmatrix} a_1(i) & \frac{a_4(i)}{2} & \frac{a_6(i)}{2} \\ \frac{a_4(i)}{2} & a_2(i) & \frac{a_5(i)}{2} \\ \frac{a_6(i)}{2} & \frac{a_5(i)}{2} & a_3(i) \end{pmatrix}, B(i)=\begin{pmatrix} a_7(i) \\ a_8(i) \\ a_9(i) \end{pmatrix}, C(i)=a_{10}(i),$$

respectively. Applying lemma 3.2, we have

$$\begin{aligned} &\int_{-\infty}^{b}\int_{-\infty}^{b'}\int_{k''}^{b''} J(i)dxdydz \\ =\ & E(i)[H(b'',b',b,A(i),B(i),C(i))-H(k'',b',b,A(i),B(i),C(i))], \quad \forall i=2,3,\cdots,8, \end{aligned}$$

where

$$
\begin{aligned}
a_1(i) &= -\frac{1}{2\,(T-t_2)}, \\
a_2(i) &= -\left(\frac{k_2^2}{2\,(T-t_2)} + \frac{1}{2(t_2-t_1)}\right), \\
a_3(i) &= -\left(\frac{k_1^2}{2\,(t_2-t_1)} + \frac{1}{2t_1}\right), \\
a_6(i) &= 0, \\
a_4(i) &= \begin{cases} \frac{k_2}{T-t_2} & \text{if } i = 4l+1 \text{ or } i = 4l+2 \\ -\frac{k_2}{T-t_2} & \text{otherwise} \end{cases}, \\
a_5(i) &= \begin{cases} \frac{k_2}{t_2-t_1} & \text{if } i = 8l+1,\ 8l+2,\ 8l+3, \text{ or } 8l+4 \\ -\frac{k_2}{t_2-t_1} & \text{otherwise} \end{cases}, \quad \forall i = 1, 2, \cdots, 16 \\
E(2) &= E(3) = E(5) = E(8) = \frac{K}{\sqrt{8\pi^3 (T-t_2)(t_2-t_1)t_1}}, \\
E(1) &= E(4) = E(6) = E(7) = -E(2), \\
E(9) &= E(12) = E(14) = E(15) = \frac{S''}{\sqrt{8\pi^3 (T-t_2)(t_2-t_1)t_1}}, \\
E(10) &= E(11) = E(13) = E(16) = -E(9),
\end{aligned}
$$

and the rest parameters are given by the following table:

i	$a_7(i)$	$a_8(i)$	$a_9(i)$	$a_{10}(i)$
1	α	$\alpha-\alpha k_2$	$\alpha-\alpha k_1$	$-\frac{T\alpha^2}{2}$
2	α	$\alpha-\alpha k_2$	$\frac{2b}{t_1}+\alpha-\alpha k_1$	$-\frac{2b^2}{t_1}-\frac{T\alpha^2}{2}$
3	$\frac{2b''}{T-t_2}+\alpha$	$\alpha+k_2\left(\frac{2b''}{T-t_2}-\alpha\right)$	$\alpha-\alpha k_1$	$-\frac{4b''^2+T^2\alpha^2-T\alpha^2t_2}{2T-2t_2}$
4	$\frac{2b''}{T-t_2}+\alpha$	$\alpha+k_2\left(\frac{2b''}{T-t_2}-\alpha\right)$	$\frac{2b}{t_1}+\alpha-\alpha k_1$	$\frac{4(t_2-T)b^2+t_1\left(-4b''^2-T^2\alpha^2+T\alpha^2t_2\right)}{2t_1(T-t_2)}$
5	α	$\frac{2b'}{t_2-t_1}+\alpha-\alpha k_2$	$\alpha+k_1\left(\frac{2b'}{t_2-t_1}-\alpha\right)$	$\frac{4b'^2-T\alpha^2t_1+T\alpha^2t_2}{2t_1-2t_2}$
6	α	$\frac{2b'}{t_2-t_1}+\alpha-\alpha k_2$	$\frac{2b}{t_1}+\alpha+k_1\left(\frac{2b'}{t_2-t_1}-\alpha\right)$	$\frac{4t_2b^2-T\alpha^2t_1^2+t_1\left(-4b^2+4b'^2+T\alpha^2t_2\right)}{2t_1(t_1-t_2)}$
7	$\frac{2b''}{T-t_2}+\alpha$	$\frac{2b'}{t_2-t_1}+\alpha+k_2\left(\frac{2b''}{T-t_2}-\alpha\right)$	$\alpha+k_1\left(\frac{2b'}{t_2-t_1}-\alpha\right)$	$\frac{2b'^2}{t_1-t_2}-\frac{T\alpha^2}{2}-\frac{2b''^2}{T-t_2}$
8	$\frac{2b''}{T-t_2}+\alpha$	$\frac{2b'}{t_2-t_1}+\alpha+k_2\left(\frac{2b''}{T-t_2}-\alpha\right)$	$\frac{2b}{t_1}+\alpha+k_1\left(\frac{2b'}{t_2-t_1}-\alpha\right)$	$-\frac{2b^2}{t_1}-\frac{T\alpha^2}{2}-\frac{2b''^2}{T-t_2}+\frac{2b'^2}{t_1-t_2}$
9	$\alpha+\sigma$	$\alpha-\alpha k_2$	$\alpha-\alpha k_1$	$-\frac{T\alpha^2}{2}$
10	$\alpha+\sigma$	$\alpha-\alpha k_2$	$\frac{2b}{t_1}+\alpha-\alpha k_1$	$-\frac{2b^2}{t_1}-\frac{T\alpha^2}{2}$
11	$\frac{2b''}{T-t_2}+\alpha+\sigma$	$\alpha+k_2\left(\frac{2b''}{T-t_2}-\alpha\right)$	$\alpha-\alpha k_1$	$-\frac{4b''^2+T^2\alpha^2-T\alpha^2t_2}{2T-2t_2}$
12	$\frac{2b''}{T-t_2}+\alpha+\sigma$	$\alpha+k_2\left(\frac{2b''}{T-t_2}-\alpha\right)$	$\frac{2b}{t_1}+\alpha-\alpha k_1$	$\frac{4(t_2-T)b^2+t_1\left(-4b''^2-T^2\alpha^2+T\alpha^2t_2\right)}{2t_1(T-t_2)}$
13	$\alpha+\sigma$	$\frac{2b'}{t_2-t_1}+\alpha-\alpha k_2$	$\alpha+k_1\left(\frac{2b'}{t_2-t_1}-\alpha\right)$	$\frac{4b'^2-T\alpha^2t_1+T\alpha^2t_2}{2t_1-2t_2}$
14	$\alpha+\sigma$	$\frac{2b'}{t_2-t_1}+\alpha-\alpha k_2$	$\frac{2b}{t_1}+\alpha+k_1\left(\frac{2b'}{t_2-t_1}-\alpha\right)$	$\frac{4t_2b^2-T\alpha^2t_1^2+t_1\left(-4b^2+4b'^2+T\alpha^2t_2\right)}{2t_1(t_1-t_2)}$
15	$\frac{2b''}{T-t_2}+\alpha+\sigma$	$\frac{2b'}{t_2-t_1}+\alpha+k_2\left(\frac{2b''}{T-t_2}-\alpha\right)$	$\alpha+k_1\left(\frac{2b'}{t_2-t_1}-\alpha\right)$	$\frac{2b'^2}{t_1-t_2}-\frac{T\alpha^2}{2}-\frac{2b''^2}{T-t_2}$
16	$\frac{2b''}{T-t_2}+\alpha+\sigma$	$\frac{2b'}{t_2-t_1}+\alpha+k_2\left(\frac{2b''}{T-t_2}-\alpha\right)$	$\frac{2b}{t_1}+\alpha+k_1\left(\frac{2b'}{t_2-t_1}-\alpha\right)$	$-\frac{2b^2}{t_1}-\frac{T\alpha^2}{2}-\frac{2b''S^2}{T-t_2}+\frac{2b'^2}{t_1-t_2}$

Finally, we obtain the following pricing formula in two dividend case:

$$\begin{aligned}\ddot{C} &= e^{-rT}\int_{-\infty}^{b}\int_{-\infty}^{b'}\int_{k''}^{b''} J(1)+J(2)+\cdots+J(16)dxdydz \\ &= e^{-rT}\sum_{i=1}^{16}\Big[E(i)[H(b'',b',b,A(i),B(i),C(i))-H(k'',b',b,A(i),B(i),C(i))]\Big].\end{aligned}$$

4 Numerical Results

This section gives the numerical results comparison of our formula and other pricing schemes to show that our approximating formula prices accurately.

Since there are no closed form formula for pricing a barrier option with discrete dividend, it is hard to find a benchmark and measure how accurate our pricing results are. Fortunately, Eq. (5) gives the exact option value in the special case when the first dividend $c_1 = 0$, and hence we can use Eq. (5) as a benchmark. Table 1 gives the comparison of the pricing results by our formula in single dividend case and Eq. (5), with settings $r = 0.03$, $\sigma = 0.2$, $K = 50$, $B = 65$, $T = 1$, $t_1 = 0.5$, $c_1 = 0$. Table 2 also gives the comparison, where our pricing results is given by the formula in two-dividend case, with all settings the same as Table 1, but $c_1 = 0$. As you can see, our formula price accurately.

$S(0)$	44	46	48	50	52
exact	1.00632706	1.251835	1.459959	1.601375	1.653843
ours	1.006327205	1.25184	1.459957	1.601373	1.653842
error	-1.44794E-07	-4.6E-06	1.61E-06	1.81E-06	9.21E-07

Table 1: Comparing the exact values and the pricing results of our approximation in single-dividend case
Option values in different initial underlying stock prices with $r = 0.03$, $\sigma = 0.2$, $K = 50$, $B = 65$, $T = 1$, $t_1 = 0.5$, $c_1 = 0$. The exact value is given by Eq. (5), "ours" gives the pricing results by our closed form approximating formula for single-dividend case, and "error" gives the difference between our approximation and exact value.

$S(0)$	44	46	48	50	52
exact	1.00632706	1.251835	1.459959	1.601375	1.653843
ours	1.006327205	1.25184	1.459957	1.601373	1.653842
error	-1.44794E-07	-4.6E-06	1.61E-06	1.81E-06	9.21E-07

Table 2: Comparing the exact values and the pricing results of our approximation in two-dividend case
All settings is equal to Table 1, but "ours" represent the pricing results by our closed form approximating formula for two-dividend case.

Next we shows the comparison of the pricing results with positive discrete dividend by our pricing formula and Monte Carlo simulation.

5 Conclusions

There are several different ways to model the stock price process with discrete dividend. As suggested by Frishling (2002), only model 3 can reflect real world phenomenon. However, it is hard to price a barrier option under model 3. Our paper suggests a way to derive analytically approximating pricing formulas for a barrier option under model 3. Through the resulting analytical formulas involve multiple integral, as shown above, all of them can be reformulated in terms of the CDF of a multivariate normal distribution. As a result, our pricing formula price efficiently. Furthermore, numerical results show that our pricing formulas produce accurate result.

A Solve the Integration of Exponential Functions by the CDF of Multi-Variate Normal Distribution

If the price of the underlying asset is assumed to follow the lognormal diffusion process, most option pricing formulas, including the pricing formulas in this paper, can be expressed in terms of multiple integrations of an exponential function, where the exponent term is a quadratic function of integrators x_1, x_2, $\cdots$. The integration problem can be numerically solved by reexpressing the formulas in terms of CDF of Multi-Variate Normal Distribution, which can be efficiently solved by accurate numerical approximation methods (see Hull (2003)). These numerical methods are provided by mathematical softwares, like Matlab and Mathemaica. Take the simplest case– the single integral for example. Under the premise $a_2 < 0$[4], the integral $\int_{-\infty}^{l} e^{a_2x^2+a_1x+a_0}dx$ can be rewritten as

$$
\begin{aligned}
&\int_{-\infty}^{l} e^{a_2x^2+a_1x+a_0}dx \\
=\ &\int_{-\infty}^{l} e^{-\frac{1}{2}\left(\frac{x-\mathbf{m}}{\mathbf{s}}\right)^2-\frac{a_1^2-4a_0a_2}{4a_2}}dx \\
=\ &\sqrt{2\pi}e^{-\frac{a_1^2-4a_0a_2}{4a_2}}\int_{-\infty}^{\frac{l-\mathbf{m}}{\mathbf{s}}}\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}y^2}\mathbf{s}dy \\
=\ &\sqrt{\frac{\pi}{-a_2}}e^{-\frac{a_1^2-4a_0a_2}{4a_2}}N\left(\frac{l-\mathbf{m}}{\mathbf{s}}\right),
\end{aligned}
\tag{29}
$$

where $\mathbf{m} = -\frac{a_1}{2a_2}$ and $\mathbf{s} = \frac{1}{\sqrt{-2a_2}}$ in the second equality, change of variables is applied with the equation $y = \frac{x-\mathbf{m}}{\mathbf{s}}$ in the third equality, $N(\cdot)$ in the last equality denotes the cdf of a univariate standard normal distribution. Note that the above identity is used to derive the pricing formula in no-dividend case in section 2.1.

However, the integration for multivariate case is not straightforward. To address this problem, we derive a general formula for the multivariate integration with n integrators: $x_1, x_2, \cdots, x_n$. Some matrix and vector calculations are employed to simplify derivation. For simplicity, for any matrix $\daleth$, we use $|\daleth|$, $\daleth^T$ and $\daleth^{-1}$ to denote the determinant, the transpose, and the inverse matrix of $\daleth$. $\daleth_{i,j}$ stands for the element located at the i-th row and j-th column of $\daleth$. For any vector ν, we use ν_i to denote the i-th element of ν. We further assume that $\mathbf{x} = (x_1, x_2, \cdots, x_n)^T$ is a column vector with n variables, B is an $n \times 1$ constant vector, C is a constant, and A is an $n \times n$ symmetric invertible negative-definite constant matrix. Then the general integral formula is derived in the following theorem:

[4] The definite integral is infinite if $a_2 > 0$.

Theorem A.1 *For any general quadratic formula $\mathbf{x}^T A\mathbf{x} + B^T\mathbf{x} + C$, the n-variate integral for $e^{\mathbf{x}^T A\mathbf{x}+B^T\mathbf{x}+C}$, that is*

$$\int_{-\infty}^{p_n}\int_{-\infty}^{p_{n-1}}\cdots\int_{-\infty}^{p_1} e^{\mathbf{x}^T A\mathbf{x}+B^T\mathbf{x}+C} d\mathbf{x} \tag{30}$$

can be expressed in terms of a cdf of a n-dimensional standard normal distribution

$$F_{Y_1,Y_2,\cdots,Y_n}(l_1, l_2, \cdots, l_n, \Sigma) \equiv \int_{-\infty}^{l_n}\int_{-\infty}^{l_{n-1}}\cdots\int_{-\infty}^{l_1} \frac{1}{\sqrt{|\Sigma|(2\pi)^n}} e^{-\frac{1}{2}\mathbf{y}^T\Sigma^{-1}\mathbf{y}} d\mathbf{y},$$

where Σ denotes the covariance matrix of n-variate standard normal random normal variables $Y_1, Y_2, \cdots, Y_n$.

Proof.

To express the integral in Eq. (30) in terms of a cdf of a standard normal distribution, the exponent term $\mathbf{x}^T A\mathbf{x} + B^T\mathbf{x} + C$ should be expressed in terms of the exponent term of a standard normal distribution. That is,

$$\mathbf{x}^T A\mathbf{x} + B^T\mathbf{x} + C = -\frac{1}{2}\mathbf{y}^T\Sigma^{-1}\mathbf{y} + C' \tag{31}$$

for some constant C'. This can be achieved by first deriving a proper constant vector $\mathbf{m}$ and a proper diagonal matrix $\mathbf{S}$ and then substituting

$$\mathbf{y} = \mathbf{S}^{-1}(\mathbf{x} - \mathbf{m}) \tag{32}$$

into the left-hand side of Eq. (31). The following lemma derive a proper $\mathbf{m}$ by completing the square identity for $\mathbf{x}^T A\mathbf{x} + B^T\mathbf{x}$.

Lemma A.2 *Under the premises that A is a symmetric invertible $n \times n$ matrix, and $\mathbf{x}, B$ are both $n \times 1$ vectors, we have*

$$\mathbf{x}^T A\mathbf{x} + B^T\mathbf{x} = \left(\mathbf{x} + \frac{1}{2}A^{-1}B\right)^T A \left(\mathbf{x} + \frac{1}{2}A^{-1}B\right) - \frac{1}{4}B^T A^{-1}B. \tag{33}$$

Proof. By expanding the right-hand side of Eq. (33), we have

$$\begin{aligned}
& \left(\mathbf{x} + \frac{1}{2}A^{-1}B\right)^T A \left(\mathbf{x} + \frac{1}{2}A^{-1}B\right) - \frac{1}{4}B^T A^{-1}B \\
= \ & \mathbf{x}^T A\mathbf{x} + \frac{1}{2}B^T\left(A^{-1}\right)^T A\mathbf{x} + \frac{1}{2}\mathbf{x}^T AA^{-1}B + \frac{1}{4}B^T\left(A^{-1}\right)^T AA^{-1}B - \frac{1}{4}B^T A^{-1}B \\
= \ & \mathbf{x}^T A\mathbf{x} + \frac{1}{2}B^T\mathbf{x} + \frac{1}{2}\mathbf{x}^T B + \frac{1}{4}B^T A^{-1}B - \frac{1}{4}B^T A^{-1}B \qquad (34) \\
= \ & \mathbf{x}^T A\mathbf{x} + B^T\mathbf{x} = \text{the left-hand side of Eq. (33)}, \qquad (35)
\end{aligned}$$

where the proposition $\left(A^{-1}\right)^T = A^{-1}$ due to the symmetry of A is substituted into Eq. (34). Since $B^T\mathbf{x}$ is a scalar, we have $B^T\mathbf{x} = \left(B^T\mathbf{x}\right)^T = \mathbf{x}^T B$. Eq. (35) is obtained by substituting the aforementioned equalities into Eq. (34). $\square$

With lemma A.1, we obtain

$$\mathbf{x}^T A\mathbf{x} + \mathbf{x}^T B + C = (\mathbf{x} - \mathbf{m})^T A(\mathbf{x} - \mathbf{m}) + C', \tag{36}$$

where

$$\mathbf{m} \equiv -\frac{1}{2}A^{-1}B, \tag{37}$$

$$C' \equiv C - \frac{1}{4}B^T A^{-1}B. \tag{38}$$

The diagonal matrix $\mathbf{S}$ can be derived by equating the right-hand sides of Eq. (31) and Eq. (36) to get

$$(\mathbf{x} - \mathbf{m})^T A(\mathbf{x} - \mathbf{m}) + C' = -\frac{1}{2}\mathbf{y}^T\Sigma^{-1}\mathbf{y} + C'.$$

Subtracting C' from both sides of above equation yields

$$(\mathbf{x} - \mathbf{m})^T A(\mathbf{x} - \mathbf{m}) = -\frac{1}{2}\mathbf{y}^T\Sigma^{-1}\mathbf{y} \tag{39}$$

$$= -\frac{1}{2}\left(\mathbf{S}^{-1}(\mathbf{x} - \mathbf{m})\right)^T \Sigma^{-1}\left(\mathbf{S}^{-1}(\mathbf{x} - \mathbf{m})\right) \tag{40}$$

$$= -\frac{1}{2}(\mathbf{x} - \mathbf{m})^T\mathbf{S}^{-1}\Sigma^{-1}\mathbf{S}^{-1}(\mathbf{x} - \mathbf{m}) \tag{41}$$

$$= -\frac{1}{2}(\mathbf{x} - \mathbf{m})^T(\mathbf{S}\Sigma\mathbf{S})^{-1}(\mathbf{x} - \mathbf{m}), \tag{42}$$

where Eq. (32) is substituted into the right-hand side of Eq. (39) to yield Eq. (40), $\left(\mathbf{S}^{-1}\right)^T = \mathbf{S}^{-1}$ due to the symmetry of $\mathbf{S}$ is substituted into Eq. (41). By comparing the right-hand side of Eq. (39) and Eq. (42), we have $-\frac{1}{2}(\mathbf{S}\Sigma\mathbf{S})^{-1} = A$, which can be rewritten as $\mathbf{S}\Sigma\mathbf{S} = (-2A)^{-1}$. Recall that $\mathbf{S}$ is a diagonal matrix. All diagonal elements of Σ are all 1 since Σ is a covariance matrix of multivariate standard normal random variables. Thus we have $(\mathbf{S}\Sigma\mathbf{S})_{i,i} = \mathbf{S}^2_{i,i}$, which leads us to obtain

$$\mathbf{S}_{i,j} \equiv \begin{cases} \sqrt{((-2A)^{-1})_{i,i}} & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}, \tag{43}$$

and

$$\Sigma \equiv (-2\mathbf{S}A\mathbf{S})^{-1}. \tag{44}$$

Now we can evaluate Eq. (30) with C', $\mathbf{m}$, $\mathbf{S}$ and Σ defined above. By applying the change of variable Eq. (32), Eq. (30) can be rewritten as

$$\begin{aligned} & \int_{x_n=-\infty}^{x_n=p_n} \int_{x_{n-1}=-\infty}^{x_{n-1}=p_{n-1}} \cdots \int_{x_1=-\infty}^{x_1=p_1} e^{\mathbf{x}^T A\mathbf{x}+B^T\mathbf{x}+C} d\mathbf{x} \\ = & \int_{x_n=-\infty}^{x_n=p_n} \int_{x_{n-1}=-\infty}^{x_{n-1}=p_{n-1}} \cdots \int_{x_1=-\infty}^{x_1=p_1} e^{-\frac{1}{2}\mathbf{y}^T\Sigma^{-1}\mathbf{y}+C'} \left|\frac{\partial \mathbf{x}}{\partial \mathbf{y}}\right| d\mathbf{y}. \end{aligned} \tag{45}$$

Since the elements in vector $\mathbf{y}$ can be represented as $\left(\frac{x_1-\mathbf{m}_1}{\mathbf{S}_{1,1}}, \frac{x_2-\mathbf{m}_2}{\mathbf{S}_{2,2}}, \cdots, \frac{x_n-\mathbf{m}_n}{\mathbf{S}_{n,n}}\right)^T$, the Jacobian determinant can be straightforwardly computed to get $\left|\frac{\partial \mathbf{x}}{\partial \mathbf{y}}\right| = \prod_{i=1}^n \mathbf{S}_{i,i} = |\mathbf{S}|$. Thus, Eq. (45) can be further rewritten as the following closed form formula:

$$\begin{aligned} & e^{C'}|\mathbf{S}| \int_{-\infty}^{\frac{p_n-\mathbf{m}_n}{\mathbf{S}_{n,n}}} \int_{-\infty}^{\frac{p_{n-1}-\mathbf{m}_{n-1}}{\mathbf{S}_{n-1,n-1}}} \cdots \int_{-\infty}^{\frac{p_1-\mathbf{m}_1}{\mathbf{S}_{1,1}}} e^{-\frac{1}{2}\mathbf{y}^T\Sigma^{-1}\mathbf{y}} d\mathbf{y} \\ = & e^{C'}|\mathbf{S}|\sqrt{|\Sigma|}\sqrt{(2\pi)^n} \int_{-\infty}^{\frac{p_n-\mathbf{m}_n}{\mathbf{S}_{n,n}}} \int_{-\infty}^{\frac{p_{n-1}-\mathbf{m}_{n-1}}{\mathbf{S}_{n-1,n-1}}} \cdots \int_{-\infty}^{\frac{p_1-\mathbf{m}_1}{\mathbf{S}_{1,1}}} \frac{1}{\sqrt{|\Sigma|(2\pi)^n}} e^{-\frac{1}{2}\mathbf{y}^T\Sigma^{-1}\mathbf{y}} d\mathbf{y} \\ = & e^{C'}\sqrt{\frac{\pi^n}{|-A|}} F_{Y_1,Y_2,\cdots,Y_n}\left(\frac{p_1-\mathbf{m}_1}{\mathbf{S}_{1,1}}, \frac{p_2-\mathbf{m}_2}{\mathbf{S}_{2,2}}, \cdots, \frac{p_n-\mathbf{m}_n}{\mathbf{S}_{n,n}}, \Sigma\right), \end{aligned} \tag{46}$$

where $|\mathbf{S}|\sqrt{|\Sigma|} = \sqrt{|\mathbf{S}\Sigma\mathbf{S}|} = \sqrt{|-2A|^{-1}}$, and $|-2A| = 2^n|-A|$ are substituted into Eq. (46). Note that the univariate integration in Eq. (29) is a special case of this theorem. □

Let us consider the bivariate integration case (i.e, $n = 2$) given in the following corollary, which is useful for evaluating the integrations for $I(1)$, $I(2)$, $\cdots$, $I(8)$ in Eq. (17) for pricing a single-dividend barrier option.

Corollary A.3 *The bivariate integration*

$$\int_{-\infty}^{q} \int_{-\infty}^{p} e^{a_1x^2+a_2xy+a_3y^2+a_4x+a_5y+a_6} dxdy$$

can be expressed as a cdf of bivariate standard normal distribution function by Theorem A.1 as

$$\frac{2\pi}{\sqrt{\Delta}} \exp\left(a_6 + \frac{a_2a_4a_5 - a_3a_4^2 - a_1a_5^2}{\Delta}\right) F_{Y_1,Y_2}\left(\frac{\sqrt{\Delta}p + \frac{2a_3a_4-a_2a_5}{\sqrt{\Delta}}}{\sqrt{-2a_3}}, \frac{\sqrt{\Delta}q + \frac{2a_1a_5-a_2a_4}{\sqrt{\Delta}}}{\sqrt{-2a_1}}, \Sigma\right),$$

where $\Delta = 4a_1a_3 - a_2^2$.

Proof. To make $a_1x^2 + a_2xy + a_3y^2 + a_4x + a_5y + a_6$ equal $\mathbf{x}^T A\mathbf{x} + \mathbf{x}^T B + C$, we set

$$A \equiv \begin{pmatrix} a_1 & \frac{a_2}{2} \\ \frac{a_2}{2} & a_3 \end{pmatrix}, B \equiv \begin{pmatrix} a_4 \\ a_5 \end{pmatrix}, C \equiv a_6.$$

By Eqs. (37), (38), (43) and (44), we have

$$\begin{aligned}
\mathbf{m} &= -\frac{1}{2}A^{-1}B = -\frac{1}{4a_1a_3 - a_2^2}\begin{pmatrix} 2a_3a_4 - a_2a_5 \\ 2a_1a_5 - a_2a_4 \end{pmatrix}, \\
C' &= C - \frac{1}{4}B^T A^{-1}B = a_6 + \frac{a_2a_4a_5 - a_3a_4^2 - a_1a_5^2}{4a_1a_3 - a_2^2}, \\
\mathbf{S}_{1,1} &= \sqrt{((-2A)^{-1})_{1,1}} = \sqrt{\frac{-2a_3}{4a_1a_3 - a_2^2}}, \\
\mathbf{S}_{2,2} &= \sqrt{((-2A)^{-1})_{2,2}} = \sqrt{\frac{-2a_1}{4a_1a_3 - a_2^2}}, \\
\Sigma &= (-2\mathbf{S}A\mathbf{S})^{-1} = \begin{pmatrix} 1 & \frac{a_2}{2\sqrt{a_1a_3}} \\ \frac{a_2}{2\sqrt{a_1a_3}} & 1 \end{pmatrix}, \\
|-A| &= \frac{4a_1a_3 - a_2^2}{4}.
\end{aligned}$$

By substituting the above into Eq. (46), we obtain

$$\begin{aligned}
&\int_{-\infty}^{q}\int_{-\infty}^{p} e^{a_1x^2 + a_2xy + a_3y^2 + a_4x + a_5y + a_6}dxdy \\
= \;& e^{C'}\sqrt{\frac{\pi^2}{|-A|}}F_{Y_1,Y_2}\left(\frac{p - \mathbf{m}(1)}{\mathbf{S}_{1,1}}, \frac{q - \mathbf{m}(2)}{\mathbf{S}_{2,2}}, \Sigma\right) \\
= \;& \frac{2\pi}{\sqrt{\Delta}}\exp\left(a_6 + \frac{a_2a_4a_5 - a_3a_4^2 - a_1a_5^2}{\Delta}\right)F_{Y_1,Y_2}\left(\frac{\sqrt{\Delta}p + \frac{2a_3a_4 - a_2a_5}{\sqrt{\Delta}}}{\sqrt{-2a_3}}, \frac{\sqrt{\Delta}q + \frac{2a_1a_5 - a_2a_4}{\sqrt{\Delta}}}{\sqrt{-2a_1}}, \Sigma\right)
\end{aligned}$$

where $\Delta = 4a_1a_3 - a_2^2$. $\square$

References

Barone-Adesi, G. and R. Whaley (1987). Efficient Analytic Approximation of American Options Values. *Journal of Finance 42*, 301–20.

Black, F. (1975). Fact and Fantasy in the Use of Options. *Financial Analysts Journal 31*, 36–41, 61–72.

Broadie, M. and J. Detemple (1995). American Capped Call Options on Dividend Paying Assets. *Review of Financial Studies 8*, 161–91.

Broadie, M. and J. Detemple (1996). American Options Valuation: New Bounds. Approximations and a Comparison of Existing Methods. *Review of Financial Studies 9*, 1211–50.

Chang, S.-L. and M. Shackleton (2003). The Simplest American and Real Option Approximations: Geske–Johnson Interpolation in Maturity and Yield. *Applied Economics Letters 10*, 709–16.

Chiras, D. and S. Manaster (1978). The Informational Content of Option Prices and a Test of Market Efficiency. *Journal of Financial Economics 6*, 213–34.

Dai, T.-S. (2009). Efficient Option Pricing on Stocks Paying Discrete or Path-Dependent Dividends with the Stair Tree. *Quantitative Finance 9*, 827–838.

Dai, T.-S. and Y.-D. Lyuu (2009). Accurate Approximation Formulas for Stock Options with Discrete Dividends. *Applied Economics Letters*, 1657–63.

Frishling, V. (2002). A Discrete Question. *Risk*, 115–6.

Heath, D. and R. Jarrow (1988). Exdividend Stock Price Behaviour and Arbitrage Opportunities. *Journal of Business 61*, 95–108.

Hull, J. (2003). Options, Futures, and Other Derivatives. pp. 266.

Krausz, J. (1985). Option Parameter Analysis and Market Efficiency Tests: a Simultaneous Solution Approach. *Applied Economics 17*, 885–96.

Merton, R. (1973). Theory of Rational Option Pricing. *Journal of Economics and Management Science 4*, 141–83.

Musiela, M. and M. Rutkowski (1997). Martingale Methods in Financial Modeling.

Roll, R. (1977). An Analytic Valuation Formula for Unprotected American Call Options on Stocks with Known Dividends. *Journal of Financial Economics 5*, 251–8.

Rubinstein, M. and E. Reiner (1991). Breaking Down the Barriers. *Risk 4*, 28–35.

Shackleton, M. and R. Wojakowski (2001). On the Expected Payoff and True Probability of Exercise of European Options. *Applied Economics Letters 8*, 269–71.